Unit 4. Regression Diagnostics

1 The Iterative Approach

FREES §5.1, §5.3 (opening), §5.10.1 ·ISLR §3.3.3 (opening). SRM learning outcomes 2(a), 2(d).

Remark (The classical assumptions; Unit 2)**.** A1–A5 are Unit 2's classical assumptions on $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ fixed: linearity (A1), zero-mean errors (A2), $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ with $\sigma^2 > 0$ (A3), $\text{rank}(\mathbf{X}) = k+1$, with $n > k+1$ for inference (A4), and normal errors (A5).

Definition 4.1 (Iterative model specification; FREES §5.1)**.** A model is specified by iterating three stages: **formulation**, from graphs of the data and prior knowledge of the relationships; **fitting**, under the assumptions of the formulated model; and **diagnostic checking**, or data and model criticism, which compares the fitted model with the data and returns the analyst to formulation when they disagree. Inferences are drawn from the model only after the data and the model agree.

Proposition 4.2 (What the residuals carry; FREES §5.3, §5.10.1)**.** Let $\mathbf{X}$ satisfy A4, and let $\mathbf{Y}$

have mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, neither $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ nor $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_n$ being assumed. Then

$$\mathsf{E}[\mathbf{e}] = \mathbf{M}\boldsymbol{\mu}, \qquad \operatorname{var}(\mathbf{e}) = \mathbf{M}\boldsymbol{\Sigma}\mathbf{M},$$

and $\mathsf{E}[\mathbf{e}] = \mathbf{0}$ iff $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$. Under A1–A3, $\operatorname{var}(\mathbf{e}) = \sigma^2\mathbf{M}$: $\mathsf{E}[e_i] = 0$ and $\operatorname{var}(e_i) = \sigma^2(1 - h_{ii})$.

Proof. $\mathbf{e} = \mathbf{MY}$ with $\mathbf{M}$ constant (Unit 2's definition of the residual maker), so $\mathsf{E}[\mathbf{e}] = \mathbf{M}\boldsymbol{\mu}$, and by the prerequisites' proposition **variance under a linear map**, $\operatorname{var}(\mathbf{e}) = \mathbf{M}\boldsymbol{\Sigma}\mathbf{M}' = \mathbf{M}\boldsymbol{\Sigma}\mathbf{M}$, $\mathbf{M}$ being symmetric. If $\boldsymbol{\mu} = \mathbf{Xc}$ then $\mathbf{M}\boldsymbol{\mu} = \mathbf{MXc} = \mathbf{0}$ by Unit 2's theorem **properties of H and M**; conversely $\mathbf{M}\boldsymbol{\mu} = \mathbf{0}$ says $\boldsymbol{\mu} = \mathbf{H}\boldsymbol{\mu}$, a vector of $\mathcal{C}(\mathbf{X})$. Under A1, $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$; under A3, $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_n$ and $\mathbf{M}\boldsymbol{\Sigma}\mathbf{M} = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}$, whose (i,i) entry is $\sigma^2(1 - h_{ii})$. □

Remark (FREES §5.3.1)**.** A pattern in the residuals estimates $\mathbf{M}\boldsymbol{\mu}$, the part of the mean that the explanatory variables do not reach. FREES lists four patterns: unusual residuals, residuals related to other explanatory variables, heteroscedastic residuals, and residuals related to time.

Remark (Potential problems; ISLR §3.3.3)**.** ISLR lists six problems of a fitted linear regression: non-linearity of the response–predictor relationship, a failure of A1; correlated errors and non-constant error variance, failures of A3; outliers and high leverage points, single observations unusual in y or in $\mathbf{x}$; and collinearity, a near-failure of A4.

2 Residual Analysis

FREES §5.3.1–5.3.3, §5.10.1–5.10.2 ·ISLR §3.3.3 (non-linearity, correlation of error terms, outliers). SRM learning outcome 2(d).

2.1 Three scalings of a residual

Proposition 4.3 (Deleting an observation is adding its indicator; FREES §5.3.2, §5.10.2)**.** Let $\mathbf{u}_i$ be the i-th coordinate vector of $\mathbb{R}^n$, let $\mathbf{X}_{(i)}$ and $\mathbf{Y}_{(i)}$ be $\mathbf{X}$ and $\mathbf{Y}$ without row i, and suppose A4 and $h_{ii} < 1$.

(i) $\mathbf{X}_{(i)}$ and $[\mathbf{X}\ \mathbf{u}_i]$ have full column rank, so the regression without observation i has a unique coefficient vector $\hat{\boldsymbol{\beta}}_{(i)}$, with residual sum of squares $\mathrm{SSE}_{(i)}$, and the regression of $\mathbf{Y}$ on $[\mathbf{X}\ \mathbf{u}_i]$

has a unique coefficient vector $(\tilde{\boldsymbol{\beta}}', \hat{\gamma})'$.

(ii) $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}_{(i)}$ and $\hat{\gamma} = y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}_{(i)}$, the **deleted residual** $e_{(i)}$; the residual sum of squares of the regression on $[\mathbf{X}\ \mathbf{u}_i]$ is $\mathrm{SSE}_{(i)}$.

(iii) $e_{(i)} = e_i/(1-h_{ii})$ and $\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\, e_i/(1-h_{ii})$.

(iv) $\mathrm{SSE}_{(i)} = \mathrm{SSE} - e_i^2/(1-h_{ii})$.

Proof. Step 1: rank. If $\mathbf{u}_i = \mathbf{Xc}$ then $\mathbf{Hu}_i = \mathbf{HXc} = \mathbf{Xc} = \mathbf{u}_i$ and $h_{ii} = \mathbf{u}_i'\mathbf{Hu}_i = \mathbf{u}_i'\mathbf{u}_i = 1$; so $h_{ii} < 1$ gives $\mathbf{u}_i \notin \mathcal{C}(\mathbf{X})$. If $\mathbf{Xv} + \gamma\mathbf{u}_i = \mathbf{0}$, then $\gamma \neq 0$ would put $\mathbf{u}_i = -\mathbf{Xv}/\gamma$ in $\mathcal{C}(\mathbf{X})$, so $\gamma = 0$, $\mathbf{Xv} = \mathbf{0}$ and $\mathbf{v} = \mathbf{0}$ by A4. If $\mathbf{X}_{(i)}\mathbf{v} = \mathbf{0}$, then $\mathbf{Xv} = (\mathbf{x}_i'\mathbf{v})\mathbf{u}_i$ lies in $\mathcal{C}(\mathbf{X})$, which forces $\mathbf{x}_i'\mathbf{v} = 0$, so $\mathbf{Xv} = \mathbf{0}$ and $\mathbf{v} = \mathbf{0}$. Unit 2's theorem **the least squares estimator** gives the unique coefficient vectors.

Step 2: the two regressions coincide. For every $\boldsymbol{\beta}$ and γ,

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} - \gamma\mathbf{u}_i\|^2 = \sum_{j\neq i}(y_j - \mathbf{x}_j'\boldsymbol{\beta})^2 + (y_i - \mathbf{x}_i'\boldsymbol{\beta} - \gamma)^2 \geqslant \|\mathbf{Y}_{(i)} - \mathbf{X}_{(i)}\boldsymbol{\beta}\|^2 \geqslant \mathrm{SSE}_{(i)},$$

with equality throughout at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_{(i)}$, $\gamma = y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}_{(i)}$. This pair therefore minimises the left side,

and by uniqueness it is $(\tilde{\boldsymbol{\beta}}, \hat{\gamma})$, with minimum $\mathrm{SSE}_{(i)}$: this is (ii).

Step 3: the formulas. Since $\mathbf{X}'\mathbf{u}_i = \mathbf{x}_i$, $\mathbf{u}_i'\mathbf{u}_i = 1$ and $\mathbf{u}_i'\mathbf{Y} = y_i$, the normal equations of the regression on $[\mathbf{X}\ \mathbf{u}_i]$ are

$$\mathbf{X}'\mathbf{X}\tilde{\boldsymbol{\beta}} + \mathbf{x}_i\hat{\gamma} = \mathbf{X}'\mathbf{Y}, \qquad \mathbf{x}_i'\tilde{\boldsymbol{\beta}} + \hat{\gamma} = y_i.$$

The first gives $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\hat{\gamma}$; substituting in the second, $\hat{y}_i - h_{ii}\hat{\gamma} + \hat{\gamma} = y_i$, so $\hat{\gamma}(1-h_{ii}) = e_i$. With Step 2 this is (iii). Unit 2's lemma **adding one explanatory variable**, with $\mathbf{z} = \mathbf{u}_i \notin \mathcal{C}(\mathbf{X})$, $\mathbf{z}'\mathbf{M}\mathbf{Y} = e_i$ and $\mathbf{z}'\mathbf{M}\mathbf{z} = 1 - h_{ii}$ (the (i,i) entry of $\mathbf{M}$), gives $\mathrm{SSE} - \mathrm{SSE}_{(i)} = e_i^2/(1-h_{ii})$, which is (iv). □

Remark (FREES §5.10.2, ISLR §5.1.2)**.** Part (iii) at observation i is Unit 1's LOOCV shortcut, and part (iv) is FREES's formula (5.21); FREES derives both from the prerequisites' Sherman–Morrison lemma applied to $\mathbf{X}_{(i)}'\mathbf{X}_{(i)} = \mathbf{X}'\mathbf{X} - \mathbf{x}_i\mathbf{x}_i'$.

Definition 4.4 (Standardised and studentised residuals; FREES §5.3.1, ISLR §3.3.3)**.** Let $h_{ii} < 1$, $n > k+2$, and let $s_{(i)}^2 = \mathrm{SSE}_{(i)}/(n-k-2)$ be the residual variance of the regression without

observation i (Proposition 4.3). Three scalings of the residual e_i are

$$\text{(a) } \frac{e_i}{s}, \qquad \text{(b) } r_i = \frac{e_i}{s\sqrt{1-h_{ii}}}, \qquad \text{(c) } t_i = \frac{e_i}{s_{(i)}\sqrt{1-h_{ii}}},$$

defined when $s > 0$ and $s_{(i)} > 0$. r_i is the **standardised residual** of Unit 2, the residual over the estimate of its standard deviation $\sigma\sqrt{1-h_{ii}}$ (Proposition 4.2); t_i, the **studentised residual**, estimates σ without observation i. R returns r_i as `rstandard` and t_i as `rstudent`.

Remark (FREES §5.3.1)**.** FREES motivates (a) by the spread of the residuals: with an intercept they average 0 (Unit 2's residual identities), so their sample variance is $\text{SSE}/(n-1) = s^2(n-k-1)/(n-1)$, close to s^2 when k is small against n.

Theorem 4.5 (The studentised residual; FREES §5.3.1, §5.10.2)**.** Let $h_{ii} < 1$, $n > k+2$ and $s > 0$.

(i) $\text{SSE}_{(i)} = s^2(n-k-1-r_i^2)$, so $s^2_{(i)} = s^2(n-k-1-r_i^2)/(n-k-2)$ and $r_i^2 \leqslant n-k-1$, with equality iff $\text{SSE}_{(i)} = 0$.

(ii) If $s_{(i)} > 0$,

$$t_i = r_i \sqrt{\frac{n-k-2}{n-k-1-r_i^2}}, \tag{1}$$

a strictly increasing function of r_i, and $|t_i| > |r_i|$ iff $|r_i| > 1$.

(iii) t_i is the t ratio of $\hat{\gamma}$ in the regression of $\mathbf{Y}$ on $[\mathbf{X}\ \mathbf{u}_i]$; under A1–A5, $t_i \sim t_{n-k-2}$.

(iv) If $s_{(i)} > 0$ and the other responses are held fixed, then $|t_i| \to \infty$ and $r_i^2 \to n-k-1$ as $|y_i| \to \infty$.

Proof. **(i)** By the definition of r_i, $e_i^2/(1-h_{ii}) = r_i^2 s^2$, and $\mathrm{SSE} = (n-k-1)s^2$, so Proposition 4.3(iv) reads $\mathrm{SSE}_{(i)} = s^2(n-k-1-r_i^2)$. Dividing by $n-k-2$ gives $s_{(i)}^2$, and $\mathrm{SSE}_{(i)} \geqslant 0$ gives the bound, with equality iff $\mathrm{SSE}_{(i)} = 0$.

(ii) $t_i = r_i\, s/s_{(i)}$, and by (i) $s/s_{(i)} = \sqrt{(n-k-2)/(n-k-1-r_i^2)}$. Write $m = n-k-1$ and $f(r) = r\sqrt{(m-1)/(m-r^2)}$ on $(-\sqrt{m}, \sqrt{m})$; then

$$f'(r) = \sqrt{m-1}\left\{(m-r^2)^{-1/2} + r^2(m-r^2)^{-3/2}\right\} = \frac{m\sqrt{m-1}}{(m-r^2)^{3/2}} > 0.$$

For $r_i \neq 0$, $|t_i|/|r_i| = \sqrt{(m-1)/(m-r_i^2)}$ exceeds 1 iff $m-1 > m-r_i^2$, that is iff $r_i^2 > 1$.
(iii) By Proposition 4.3, the regression on $[\mathbf{X}\ \mathbf{u}_i]$ has $\hat\gamma = e_i/(1-h_{ii})$ and residual sum of squares $\mathrm{SSE}_{(i)}$ on $n-(k+2)$ degrees of freedom, so its residual variance is $s_{(i)}^2$, and Unit 2's lemma **adding one explanatory variable** gives $\mathrm{SE}(\hat\gamma) = s_{(i)}/\sqrt{\mathbf{u}_i'\mathbf{M}\mathbf{u}_i} = s_{(i)}/\sqrt{1-h_{ii}}$. Hence $\hat\gamma/\mathrm{SE}(\hat\gamma) = e_i/(s_{(i)}\sqrt{1-h_{ii}}) = t_i$. Under A1–A5, $\mathbf{Y} = \mathbf{X}\boldsymbol\beta + 0\cdot\mathbf{u}_i + \boldsymbol\varepsilon$ satisfies A1–A5 for the design $[\mathbf{X}\ \mathbf{u}_i]$, which has full column rank $k+2<n$ by Proposition 4.3(i). Unit 2's theorem **the t statistic** with $\mathbf{c}$ the last coordinate vector and true value $\gamma = 0$ gives $t_i \sim t_{n-k-2}$.
(iv) $\mathbf{e} = \mathbf{M}\mathbf{Y}$, so $e_i = (1-h_{ii})y_i - \sum_{j\neq i} h_{ij}y_j$ is affine in y_i with slope $1-h_{ii} > 0$, and $|e_i| \to \infty$. $\mathrm{SSE}_{(i)}$ and $s_{(i)}$ do not involve y_i, so $|t_i| \to \infty$. By Proposition 4.3(iv), $\mathrm{SSE} = \mathrm{SSE}_{(i)} + E_i$ with $E_i = e_i^2/(1-h_{ii})$, and $r_i^2 = E_i/s^2 = (n-k-1)E_i/(\mathrm{SSE}_{(i)} + E_i) \to n-k-1$ as $E_i \to \infty$. □

Remark (FREES §5.3.1, ISLR §3.3.3)**.** Part (iv) is FREES's argument for t_i: an unusually large response inflates s together with e_i, so $|r_i|$ can never pass $\sqrt{n-k-1}$, while $s_{(i)}$ is untouched and $|t_i|$ grows without bound. FREES gives t_i the t law on $n-(k+1)$ degrees of freedom; part (iii) shows the count is $n-(k+1)-1$, since $s_{(i)}$ comes from $n-1$ observations and $k+1$ coefficients. ISLR defines the studentised residual as the residual over its estimated standard error,

which describes r_i and t_i alike; its lab computes t_i.

Corollary 4.6 (The largest studentised residual (beyond the syllabus))**.** Suppose A1–A5, $n > k+2$ and $h_{ii} < 1$ for every i, and let $c = t_{n-k-2,\,1-\alpha/(2n)}$. Then $\mathsf{P}\{\max_i |t_i| > c\} \leqslant \alpha$.

Proof. $\{\max_i |t_i| > c\} = \bigcup_{i=1}^{n} \{|t_i| > c\}$. By subadditivity of probability and Theorem 4.5(iii), with the t law symmetric, $\mathsf{P}\{\max_i |t_i| > c\} \leqslant \sum_{i=1}^{n} \mathsf{P}\{|t_i| > c\} = n \cdot 2 \cdot \alpha/(2n) = \alpha$. No independence among the t_i is used. □

Example 4.1 (Outliers and high leverage points, continued; FREES §5.3.1)**.** The 19 base points of Unit 2's example with one of the unusual points A, B, C added, so $n = 20$ and $k = 1$:

```
ol <- read.csv("data/frees/csv/OutlierExample.csv")
abc <- sapply(1:3, function(code) {
  d <- subset(ol, CODES %in% c(0, code)); m <- lm(Y ~ X, data = d); i <- nrow(d)
  r <- rstandard(m)[[i]]
  c(r = r, t = rstudent(m)[[i]], t_from_r = r * sqrt((20 - 1 - 2) / (20 - 1 - 1 -
   ↪ r^2)),
    s = summary(m)$sigma, s_without = summary(lm(Y ~ X, data = d[-i, ]))$sigma)
```

```
})
colnames(abc) <- c("A", "B", "C"); round(abc, 4)
```

```
##                A      B        C
## r         4.0030 0.7711  -4.0139
## t        11.7415 0.7621 -12.0424
## t_from_r 11.7415 0.7621 -12.0424
## s         0.8455 0.2849   0.8648
## s_without 0.2883 0.2883   0.2883
```

```
c(ceiling_r = sqrt(20 - 1 - 1), bonferroni_cut = qt(1 - 0.05 / (2 * 20), 20 - 1 -
↪  2))
```

```
##      ceiling_r bonferroni_cut
##          4.243          3.543
```

A and C have $r = 4.00$ and -4.01, close to the ceiling $\sqrt{n-k-1} = 4.24$ of Theorem 4.5(i). Without them the fit to the base points has $s_{(i)} = 0.288$ against $s = 0.846$ and 0.865 with them, so their studentised residuals are 11.74 and -12.04, far beyond the cutoff 3.54 of Corollary 4.6 at $\alpha = 0.05$. B has $|r| < 1$, and its t is smaller still, as Theorem 4.5(ii) requires.

2.2 Outliers

Proposition 4.7 (How often a correct model is flagged; FREES §5.3.2)**.** Under A1–A5 with $n > k + 5$ and $h_{jj} < 1$ for every j,

$$\mathsf{P}\{|r_i| > 2\} = \mathsf{P}\left\{|T| > 2\sqrt{\tfrac{n-k-2}{n-k-5}}\right\}, \qquad T \sim t_{n-k-2},$$

and the expected number of observations with $|r_i| > 2$ is n times this probability.

Proof. By Theorem 4.5(ii), $r \mapsto t$ is strictly increasing and odd, so $|r_i| > 2$ iff $|t_i| > 2\sqrt{(n-k-2)/(n-k-1-4)}$; Theorem 4.5(iii) gives the law of t_i. The count is $\sum_i \mathbb{1}\{|r_i| > 2\}$, whose expectation is the sum of the n equal probabilities. □

```
flag <- function(n, k) 2 * pt(-2 * sqrt((n - k - 2) / (n - k - 5)), n - k - 2)
round(c(n20_k1 = flag(20, 1), n123_k2 = flag(123, 2), n400_k2 = flag(400, 2),
        normal = 2 * pnorm(-2), expected_flags_n400 = 400 * flag(400, 2)), 4)
```

```
##           n20_k1          n123_k2          n400_k2
##           0.0416           0.0450           0.0454
```

```
##              normal expected_flags_n400
##              0.0455              18.1451
```

Remark (Flagging outliers; FREES §5.3.2, ISLR §3.3.3)**.** FREES flags observation i as an **outlier** when $|r_i| > 2$, and ISLR when its studentised residual exceeds 3 in absolute value. By Proposition 4.7 a correct model with $n = 400$ and $k = 2$ has each $|r_i| > 2$ with probability 0.045, about 18 flags expected, and a flag is a reason to investigate the observation, not evidence against the model.

Remark (Options for handling outliers; FREES §5.3.2)**.** FREES's options are to keep the observation and comment on its effect; to delete it as unrepresentative and report it separately; or to add an indicator variable for it, which formalises a special cause that later observations may share. By Proposition 4.3 and Theorem 4.5(iii), the indicator reproduces the deletion exactly: the same coefficients for the other variables, the same residuals and SSE for the other observations, and an exact fit at the outlier, whose indicator coefficient is the deleted residual and whose t ratio is t_i.

2.3 Residuals and other explanatory variables

Proposition 4.8 (Residuals against a candidate variable; FREES §5.3.3, §5.10.1)**.** Suppose A4, that the model has an intercept and SSE > 0, write $r(\cdot,\cdot)$ for a sample correlation, and let $\mathbf{v} \notin \mathcal{C}(\mathbf{X})$ be a candidate explanatory variable whose regression on $\mathbf{X}$ has coefficient of determination R_v^2.

(i) $r(\mathbf{e}, \mathbf{x}_j) = 0$ for every non-constant explanatory variable x_j in the model, and $r(\mathbf{e}, \hat{\mathbf{Y}}) = 0$ if $\hat{\mathbf{Y}}$ is not constant.

(ii) $r(\mathbf{e}, \mathbf{v}) = r(y, v \mid x_1, \ldots, x_k)\sqrt{1 - R_v^2}$, so $|r(\mathbf{e}, \mathbf{v})| \leqslant |r(y, v \mid x_1, \ldots, x_k)|$.

(iii) If the regression on $[\mathbf{X}\ \mathbf{v}]$ has positive residual sum of squares, the t ratio t_v of v in it satisfies $r(y, v \mid x_1, \ldots, x_k) = t_v/\sqrt{t_v^2 + n - k - 2}$.

Proof. For a vector $\mathbf{w}$ with $\mathbf{M}_0\mathbf{w} \neq \mathbf{0}$, where $\mathbf{M}_0 = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}'$, $r(\mathbf{e}, \mathbf{w}) = \mathbf{e}'\mathbf{M}_0\mathbf{w}/(\|\mathbf{M}_0\mathbf{e}\|\ \|\mathbf{M}_0\mathbf{w}\|)$. By Unit 2's residual identities $\mathbf{1}'\mathbf{e} = 0$, so $\mathbf{M}_0\mathbf{e} = \mathbf{e}$ and $\mathbf{e}'\mathbf{M}_0\mathbf{w} = \mathbf{e}'\mathbf{w}$.

(i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$ and $\hat{\mathbf{Y}}'\mathbf{e} = 0$ by the same identities.

(ii) In Unit 3's added variable plot of v in the regression on $[\mathbf{X}\ \mathbf{v}]$, the coordinates are $\mathbf{M}\mathbf{Y} = \mathbf{e}$ and $\mathbf{M}\mathbf{v}$, both with mean zero since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, so $r(y, v \mid x_1, \ldots, x_k) = \mathbf{e}'\mathbf{M}\mathbf{v}/(\|\mathbf{e}\|\ \|\mathbf{M}\mathbf{v}\|)$, and

$\mathbf{Mv} \neq \mathbf{0}$ because $\mathbf{v} \notin \mathcal{C}(\mathbf{X})$. Since $\mathbf{M}$ is symmetric and $\mathbf{Me} = \mathbf{M}^2\mathbf{Y} = \mathbf{e}$, $\mathbf{e}'\mathbf{v} = \mathbf{e}'\mathbf{Mv}$. Hence

$$r(\mathbf{e}, \mathbf{v}) = \frac{\mathbf{e}'\mathbf{Mv}}{\|\mathbf{e}\| \, \|\mathbf{M}_0\mathbf{v}\|} = r(y, v \mid x_1, \ldots, x_k) \, \frac{\|\mathbf{Mv}\|}{\|\mathbf{M}_0\mathbf{v}\|},$$

where $\|\mathbf{Mv}\|^2$ and $\|\mathbf{M}_0\mathbf{v}\|^2 > 0$ are the residual and total sums of squares of the regression of $\mathbf{v}$ on $\mathbf{X}$, whose ratio is $1 - R_v^2$ by Unit 2's variance decomposition.

(iii) Unit 3's proposition **partial correlation from the t ratio**, applied to the regression on $[\mathbf{X}\ \mathbf{v}]$, which has full column rank because $\mathbf{v} \notin \mathcal{C}(\mathbf{X})$, $k+1$ explanatory variables and $n-k-2$ residual degrees of freedom. □

Remark (FREES §5.3.3)**.** By (ii), the correlation of the residuals with a candidate understates the partial correlation that governs its t ratio by the factor $\sqrt{1 - R_v^2}$, severely when the candidate is nearly a combination of the variables already in the model; Unit 3's added variable plot shows the partial correlation itself. A correlation also misses curvature: residuals $\propto v^2 - \frac{2}{3}$ against $v \in \{-1, 0, 1\}$ lie on a parabola and have correlation 0, which a scatter plot reveals.

Example 4.2 (Stock market liquidity; FREES §5.3.3)**.** Trading volume `VOLUME` (millions of shares over three months) of 123 companies, with the number of transactions `NTRAN`, the average time between transactions `AVGT` (minutes), the opening price `PRICE`, the shares outstanding `SHARE` (millions), the market value `VALUE` $\approx$ `PRICE` $\times$ `SHARE`/1000 (billions of dollars) and the debt-to-equity ratio `DEBEQ`; FREES removed three companies with extreme volume or price before the analysis. FREES Tables 5.2 and 5.3:

```
liq <- read.csv("data/frees/csv/Liquidity.csv"); nrow(liq)
```

```
## [1] 123
```

```
v7 <- c("VOLUME", "AVGT", "NTRAN", "PRICE", "SHARE", "VALUE", "DEBEQ")
round(t(sapply(liq[v7], function(x)
  c(mean = mean(x), median = median(x), sd = sd(x), min = min(x), max = max(x)))),
  ↪ 3)
```

```
##             mean   median       sd     min      max
## VOLUME    13.423   11.556   10.632   0.658    64.57
## AVGT       5.441    4.284    3.853   0.590    20.77
## NTRAN   6436.252 5071.000 5309.994 999.000 36420.00
## PRICE     38.796   34.375   21.370   9.125   122.38
```

```
## SHARE    94.731   53.827  115.103   6.736   783.05
## VALUE     4.116    2.065    8.157   0.115    75.44
## DEBEQ     2.697    1.105    6.509   0.185    53.63
```

```
round(cor(liq[v7]), 3)
```

```
##         VOLUME   AVGT  NTRAN  PRICE  SHARE  VALUE  DEBEQ
## VOLUME   1.000 -0.674  0.913  0.168  0.773  0.702 -0.052
## AVGT    -0.674  1.000 -0.668 -0.128 -0.429 -0.318  0.094
## NTRAN    0.913 -0.668  1.000  0.190  0.817  0.760 -0.092
## PRICE    0.168 -0.128  0.190  1.000  0.177  0.457 -0.038
## SHARE    0.773 -0.429  0.817  0.177  1.000  0.829 -0.077
## VALUE    0.702 -0.318  0.760  0.457  0.829  1.000 -0.077
## DEBEQ   -0.052  0.094 -0.092 -0.038 -0.077 -0.077  1.000
```

```
c(cor_AVGT_1overNTRAN = cor(liq$AVGT, 1 / liq$NTRAN))
```

```
## cor_AVGT_1overNTRAN
##              0.9998
```

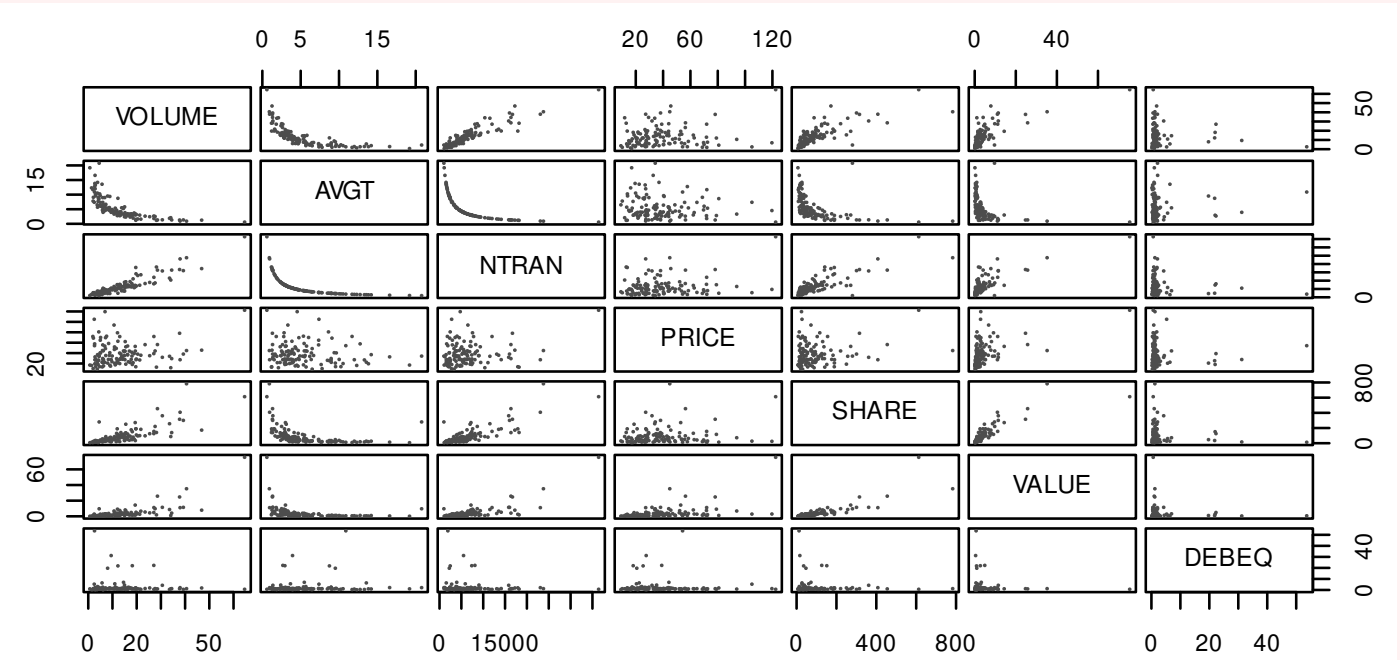


Figure 4.1: The scatterplot matrix of the seven liquidity variables (FREES Figure 5.2): the curved relation of `AVGT` to `NTRAN` and the few extreme companies are visible, which the correlation matrix alone does not show.

AVGT and 1/NTRAN have correlation 0.9998: the average time between trades times the number of trades is nearly the length of the trading period. The first fit and FREES Table 5.4, the correlations of its residuals with the other variables:

```
m1 <- lm(VOLUME ~ NTRAN, data = liq)
round(coef(summary(m1)), 6)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.651284   0.617300   2.675 0.008507
## NTRAN       0.001829   0.000074  24.680 0.000000
```

```
c(R2 = summary(m1)$r.squared, s = summary(m1)$sigma)
```

```
##     R2      s
## 0.8343 4.3463
```

```
cand <- c("AVGT", "PRICE", "SHARE", "VALUE", "DEBEQ")
round(rbind(residual = drop(cor(residuals(m1), liq[cand])),
            standardised = drop(cor(rstandard(m1), liq[cand]))), 3)
```

```
##                AVGT  PRICE SHARE VALUE DEBEQ
## residual     -0.159 -0.014 0.064 0.018 0.078
## standardised -0.155 -0.017 0.055 0.007 0.078
```

FREES's Table 5.4 prints the second row, the correlations of the standardised residuals. **AVGT** has the largest correlation, −0.159, and enters next:

```
m2 <- lm(VOLUME ~ NTRAN + AVGT, data = liq)
round(coef(summary(m2)), 6)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.408719   1.301213   3.388 0.000952
## NTRAN        0.001673   0.000098  17.134 0.000000
## AVGT        -0.322192   0.134572  -2.394 0.018205
```

```
c(R2 = summary(m2)$r.squared, s = summary(m2)$sigma)
```

```
##     R2      s
## 0.8418 4.2637
```

```
round(rbind(residual = drop(cor(residuals(m2), liq[cand[-1]])),
            standardised = drop(cor(rstandard(m2), liq[cand[-1]]))), 3)
```

```
##                PRICE SHARE VALUE DEBEQ
## residual      -0.015 0.100 0.074 0.089
## standardised -0.015 0.096 0.071 0.089
```

The fitted model is $\widehat{\texttt{VOLUME}} = 4.41 - 0.322\,\texttt{AVGT} + 0.00167\,\texttt{NTRAN}$ with $t(b_{\texttt{AVGT}}) = -2.39$, $R^2 = 84.2\%$ and $s = 4.26$, against 83.4% and 4.35 for the first model; FREES's Table 5.5 is the standardised row. Proposition 4.8(ii) links the first model's residual correlation with AVGT's t ratio in the second, and recovers the partial correlation of every remaining candidate from the second model's residuals:

```
t_A <- coef(summary(m2))["AVGT", "t value"]; n <- nrow(liq)
R2_A <- summary(lm(AVGT ~ NTRAN, data = liq))$r.squared
c(partial_from_t = t_A / sqrt(t_A^2 + n - 3), sqrt_1_minus_R2v = sqrt(1 - R2_A),
  product = t_A / sqrt(t_A^2 + n - 3) * sqrt(1 - R2_A),
  cor_resid_AVGT = cor(residuals(m1), liq$AVGT))
```

```
##   partial_from_t sqrt_1_minus_R2v          product
##          -0.2135           0.7446          -0.1590
##   cor_resid_AVGT
##          -0.1590
```

```
round(sapply(cand[-1], function(v) {
  R2v <- summary(lm(reformulate(c("NTRAN", "AVGT"), v), data = liq))$r.squared
```

```
  tv <- coef(summary(lm(reformulate(c("NTRAN", "AVGT", v), "VOLUME"), data =
   ↪  liq)))[v, 3]
  c(cor_resid = cor(residuals(m2), liq[[v]]), R2v = R2v,
    partial = cor(residuals(m2), liq[[v]]) / sqrt(1 - R2v),
    partial_from_t = tv / sqrt(tv^2 + n - 4), t = tv)
}), 4)
```

```
##                  PRICE  SHARE  VALUE  DEBEQ
## cor_resid      -0.0147 0.0998 0.0739 0.0892
## R2v             0.0362 0.6922 0.6423 0.0104
## partial        -0.0149 0.1798 0.1235 0.0897
## partial_from_t -0.0149 0.1798 0.1235 0.0897
## t              -0.1631 1.9943 1.3578 0.9824
```

SHARE and **VALUE** have residual correlations 0.100 and 0.074, but $R_v^2 = 0.69$ and 0.64 turn these into partial correlations 0.180 and 0.124, and **SHARE** would enter with $t = 1.99$. The candidates are strongly skewed:

```
skew <- function(x) mean((x - mean(x))^3) / (mean((x - mean(x))^2))^1.5
round(sapply(c(liq[c("PRICE", "SHARE", "VALUE", "DEBEQ")], list(resid =
 ↪  residuals(m2))), skew), 2)
```

```
## PRICE SHARE VALUE DEBEQ resid
##  1.47  3.18  6.26  5.42  0.60
```

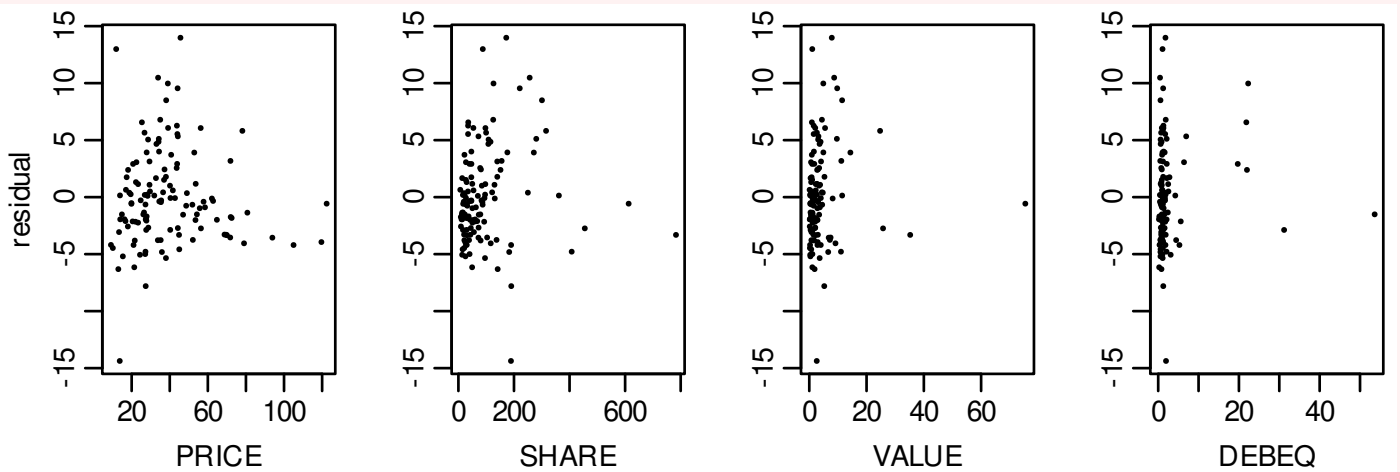


Figure 4.2: Residuals of **VOLUME** on **NTRAN** and **AVGT** against the four remaining candidates (FREES Figure 5.3).

The residuals have skewness 0.60, while **SHARE**, **VALUE** and **DEBEQ** have skewness 3.18, 6.26 and 5.42; their plots are clouds compressed against the left edge, and no plot shows a trend.

Example 4.3 (Non-linearity; ISLR §3.3.3)**.** The regression of **mpg** on **horsepower** for the 392 cars of **Auto**, with and without the square of **horsepower** (ISLR Figure 3.9):

```
nrow(Auto)
```

```
## [1] 392
```

```
a1 <- lm(mpg ~ horsepower, data = Auto)
a2 <- lm(mpg ~ horsepower + I(horsepower^2), data = Auto)
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.4, 1.4, 0.6), mgp = c(1.9, 0.6, 0))
for (m in list(a1, a2)) {
  plot(fitted(m), residuals(m), xlab = "fitted", ylab = "residual", pch = 16, cex =
  ↪ 0.4,
       main = if (identical(m, a1)) "linear" else "quadratic", cex.main = 0.9)
  lines(lowess(fitted(m), residuals(m)), col = "#B45309", lwd = 2); abline(h = 0,
  ↪ lty = 2)
}
par(op)
```

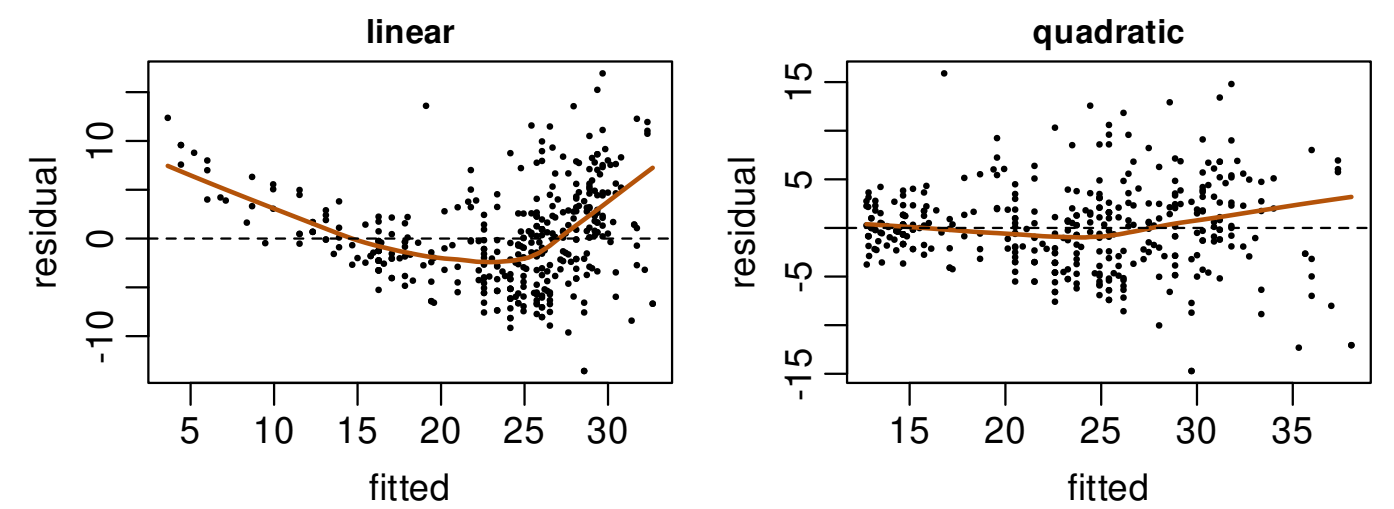


Figure 4.3: Residuals against fitted values, with a `lowess` smooth, for the linear and the quadratic fit of `mpg` on `horsepower`.

The linear fit leaves a U: positive residuals at both ends of the fitted range and negative ones in the middle, the estimate of $\mathbf{M}\boldsymbol{\mu}$ of Proposition 4.2 for a curved mean. The quadratic fit's smooth stays near 0 except at the largest fitted values. With several explanatory variables the residuals

are plotted against the fitted values, with which they have correlation 0 (Proposition 4.8(i)), so any trend in the plot is non-linear.

2.4 Correlated errors

Proposition 4.9 (Least squares under a general error covariance; ISLR §3.3.3, CLM exercise B5)**.** Assume A1, A2 and A4, and let $\mathrm{var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Sigma}$ be any covariance matrix. Then $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$,

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}, \qquad \mathsf{E}[s^2] = \frac{\mathrm{tr}(\mathbf{M}\boldsymbol{\Sigma})}{n-k-1},$$

so the reported $s^2(\mathbf{X}'\mathbf{X})^{-1}$ has expectation $\{\mathrm{tr}(\mathbf{M}\boldsymbol{\Sigma})/(n-k-1)\}(\mathbf{X}'\mathbf{X})^{-1}$.

Proof. $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ has mean $\mathbf{0}$ by A2, and the variance follows from the prerequisites' proposition **variance under a linear map**. By Unit 2's identity $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ and the symmetry and idempotency of $\mathbf{M}$, $\mathrm{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$, and the prerequisites' lemma **expectation of a quadratic form** gives $\mathsf{E}[\mathrm{SSE}] = \mathrm{tr}(\mathbf{M}\boldsymbol{\Sigma})$. Divide by $n-k-1$. □

Corollary 4.10 (Equicorrelated errors; ISLR §3.3.3)**.** In the model $Y_i = \beta_0 + \varepsilon_i$, with $k = 0$ and $\hat{\beta}_0 = \bar{Y}$, let $\mathrm{var}(\varepsilon_i) = \sigma^2$ and $\mathrm{cor}(\varepsilon_i, \varepsilon_j) = \rho$ for $i \neq j$, with $-1/(n-1) < \rho < 1$, so that the covariance matrix is positive definite. Then

$$\mathrm{var}(\bar{Y}) = \frac{\sigma^2}{n}\{1 + (n-1)\rho\}, \qquad \mathsf{E}\left[\frac{s^2}{n}\right] = \frac{\sigma^2}{n}(1-\rho),$$

so the reported variance of $\bar{Y}$ has expectation $(1-\rho)/\{1+(n-1)\rho\}$ times the true one, a factor below 1 for $\rho > 0$ that tends to 0 as $n \to \infty$.

Proof. $\boldsymbol{\Sigma} = \sigma^2\{(1-\rho)\mathbf{I}_n + \rho\mathbf{1}\mathbf{1}'\}$ and $\mathbf{X} = \mathbf{1}$, so $\mathrm{var}(\bar{Y}) = n^{-2}\mathbf{1}'\boldsymbol{\Sigma}\mathbf{1} = n^{-2}\sigma^2\{(1-\rho)n + \rho n^2\}$, as in Unit 1's proposition **mean of correlated estimates**. Here $\mathbf{M} = \mathbf{M}_0 = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}'$ with $\mathbf{M}_0\mathbf{1} = \mathbf{0}$ and $\mathrm{tr}(\mathbf{M}_0) = n - 1$, so $\mathrm{tr}(\mathbf{M}_0\boldsymbol{\Sigma}) = \sigma^2(1-\rho)(n-1)$ and Proposition 4.9 gives $\mathsf{E}[s^2] = \sigma^2(1-\rho)$. □

Proposition 4.11 (Doubled data; ISLR §3.3.3)**.** Stack the data on themselves, $\mathbf{X}_d = \begin{pmatrix}\mathbf{X}\\ \mathbf{X}\end{pmatrix}$ and

$\mathbf{Y}_d = \begin{pmatrix} \mathbf{Y} \\ \mathbf{Y} \end{pmatrix}$, and suppose A4. The least squares coefficients are unchanged, and every standard error computed from the $2n$ rows equals the original one times $\sqrt{(n-k-1)/(2n-k-1)}$, which tends to $1/\sqrt{2}$.

Proof. $\mathbf{X}_d'\mathbf{X}_d = 2\mathbf{X}'\mathbf{X}$ and $\mathbf{X}_d'\mathbf{Y}_d = 2\mathbf{X}'\mathbf{Y}$, so the normal equations are those of the original data times 2. The residual vector is $\mathbf{e}$ stacked twice, so $\mathrm{SSE}_d = 2\mathrm{SSE}$ and $s_d^2 = 2\mathrm{SSE}/(2n-k-1)$, and

$$\mathrm{SE}_d(\hat{\beta}_j) = \sqrt{\frac{2\mathrm{SSE}}{2n-k-1}}\sqrt{\frac{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}{2}} = \sqrt{\frac{n-k-1}{2n-k-1}}\sqrt{\frac{\mathrm{SSE}}{n-k-1}}\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}. \qquad \square$$

Remark (ISLR §3.3.3)**.** The duplicated errors are perfectly correlated in pairs, so $\boldsymbol{\Sigma}$ violates A3, and the standard errors behave as if there were $2n$ independent observations. Positive correlation between the errors of adjacent observations, common in time series and in clustered data, shows as **tracking** in a plot of the residuals against time: neighbouring residuals take similar values.

Example 4.4 (Tracking; ISLR §3.3.3)**.** The simple regression on $x_t = t$, $t = 1, \ldots, 100$, with errors of variance 1 and $\operatorname{cor}(\varepsilon_t, \varepsilon_u) = \rho^{|t-u|}$. The exact variance of b_1 and the exact expectation of its reported variance come from Proposition 4.9; the coverage of the nominal 95% interval for β_1 is simulated:

```
set.seed(4); nT <- 100; X <- cbind(1, 1:nT); XtXi <- solve(crossprod(X))
M <- diag(nT) - X %*% XtXi %*% t(X)
track <- t(sapply(c(0, 0.5, 0.9), function(rho) {
  S <- rho^abs(outer(1:nT, 1:nT, "-")); L <- t(chol(S))
  true_var <- (XtXi %*% t(X) %*% S %*% X %*% XtXi)[2, 2]
  rep_var <- sum(diag(M %*% S)) / (nT - 2) * XtXi[2, 2]
  cover <- mean(replicate(2000, {
    y <- 1 + 0.1 * (1:nT) + drop(L %*% rnorm(nT)); f <- lm.fit(X, y)
    se <- sqrt(sum(f$residuals^2) / (nT - 2) * XtXi[2, 2])
    abs(f$coefficients[2] - 0.1) <= qt(0.975, nT - 2) * se
  }))
  c(rho = rho, true_over_reported = true_var / rep_var, coverage = cover)
}))
round(track, 3)
```

```
##      rho true_over_reported coverage
## [1,] 0.0                1.000    0.945
## [2,] 0.5                2.998    0.749
## [3,] 0.9               19.590    0.345
```

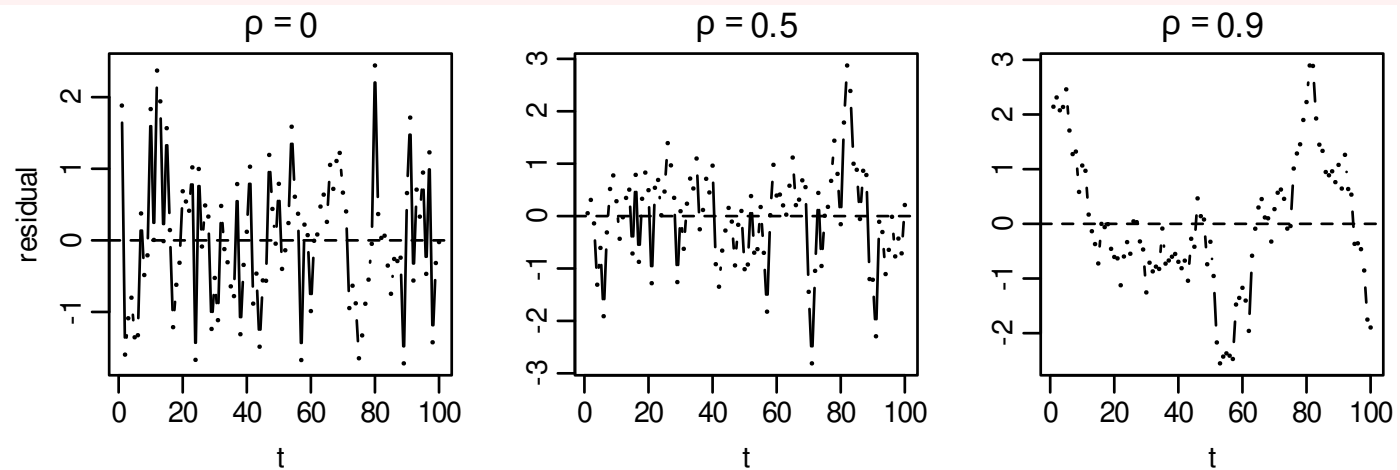


Figure 4.4: Residuals against time for $\rho = 0$, 0.5 and 0.9 (ISLR Figure 3.10).

At $\rho = 0.5$ the true variance of b_1 is 3.0 times the expectation of its reported variance and the nominal 95% interval covers β_1 in 75% of the samples; at $\rho = 0.9$ the factor is 19.6 and the coverage

35%. The residual series wander in longer runs as ρ grows.

3 Influential Points

FREES §5.4, §5.10.2 ·ISLR §3.3.3 (high leverage points). SRM learning outcome 2(d).

3.1 Leverage

Proposition 4.12 (Leverage as a distance; FREES §5.4.1, ISLR §3.3.3)**.** Suppose the model has an intercept, write $\mathbf{X} = [\mathbf{1}\ \tilde{\mathbf{X}}]$ and $\mathbf{x}_0 = (1, \tilde{\mathbf{x}}_0')'$ for a point with $\tilde{\mathbf{x}}_0 \in \mathbb{R}^k$, let $\bar{\mathbf{x}}$ be the vector of means of the k explanatory variables, $\tilde{\mathbf{X}}_c = \tilde{\mathbf{X}} - \mathbf{1}\bar{\mathbf{x}}'$, and $\mathbf{S} = \tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c/(n-1)$ their sample covariance matrix. Under A4, $\mathbf{S}$ is positive definite,

$$\mathbf{H} = \tfrac{1}{n}\mathbf{1}\mathbf{1}' + \tilde{\mathbf{X}}_c(\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c)^{-1}\tilde{\mathbf{X}}_c', \tag{2}$$

and

$$\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \frac{1}{n} + \frac{(\tilde{\mathbf{x}}_0 - \bar{\mathbf{x}})'\mathbf{S}^{-1}(\tilde{\mathbf{x}}_0 - \bar{\mathbf{x}})}{n-1}. \tag{3}$$

In particular $h_{ii} = 1/n + \Delta_i^2/(n-1)$, where $\Delta_i^2 = (\tilde{\mathbf{x}}_i - \bar{\mathbf{x}})'\mathbf{S}^{-1}(\tilde{\mathbf{x}}_i - \bar{\mathbf{x}})$ is the squared **Mahalanobis**

distance of observation i from the centre of the design; and under A1–A4 the fitted mean $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$ has variance $\sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$.

Proof. Let $\mathbf{A} = \begin{pmatrix} 1 & -\bar{\mathbf{x}}' \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}$, invertible, so that $\mathbf{XA} = [\mathbf{1} \;\; \tilde{\mathbf{X}} - \mathbf{1}\bar{\mathbf{x}}'] = [\mathbf{1} \;\; \tilde{\mathbf{X}}_c]$. The columns of $\tilde{\mathbf{X}}_c$ sum to zero, so $(\mathbf{XA})'(\mathbf{XA}) = \operatorname{diag}\big(n, \;\; \tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c\big)$. By A4 and the invertibility of $\mathbf{A}$, $\mathbf{XA}$ has full column rank, so this matrix is invertible and so is $\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c = (n-1)\mathbf{S}$; being a Gram matrix it is positive semi-definite, $\mathbf{v}'\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c\mathbf{v} = \|\tilde{\mathbf{X}}_c\mathbf{v}\|^2$, and an invertible positive semi-definite matrix is positive definite. From $(\mathbf{XA})'(\mathbf{XA}) = \mathbf{A}'\mathbf{X}'\mathbf{XA}$,

$$(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{A}\,\operatorname{diag}\big(1/n, \;\; (\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c)^{-1}\big)\mathbf{A}'.$$

Pre- and post-multiplying by $\mathbf{X}$ and $\mathbf{X}'$ gives (2), since $\mathbf{XA} = [\mathbf{1} \;\; \tilde{\mathbf{X}}_c]$. For (3), $\mathbf{A}'\mathbf{x}_0 = (1, (\tilde{\mathbf{x}}_0 - \bar{\mathbf{x}})')'$, so $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 1/n + (\tilde{\mathbf{x}}_0 - \bar{\mathbf{x}})'\{(n-1)\mathbf{S}\}^{-1}(\tilde{\mathbf{x}}_0 - \bar{\mathbf{x}})$. Taking $\mathbf{x}_0 = \mathbf{x}_i$ gives h_{ii}. Finally $\operatorname{var}(\mathbf{x}_0'\hat{\boldsymbol{\beta}}) = \mathbf{x}_0'\operatorname{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 = \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$ by Unit 2's theorem **variance–covariance matrix**. □

Remark (FREES §5.4.1, ISLR §3.3.3)**.** For $k = 1$, (3) is Unit 2's formula $h_{ii} = 1/n + (x_i - \bar{x})^2 / \sum_j (x_j - \bar{x})^2$. The distance Δ_i weighs the explanatory variables jointly, so a point can be ordinary in each variable and unusual in the design: for two variables with unit variances and correlation r, the point one standard deviation above the mean in the first and one below in the second has $\Delta^2 = (1, -1)\mathbf{S}^{-1}(1, -1)' = 2(1 + r)/(1 - r^2) = 2/(1 - r)$, which is 20 at $r = 0.9$ (FREES Figure 5.4, ISLR Figure 3.13).

Remark (High leverage points; FREES §5.4.1, ISLR §3.3.3)**.** With an intercept, Unit 2's theorem **properties of leverage** gives $1/n \leqslant h_{ii} \leqslant 1$ and the mean leverage $(k+1)/n$. FREES declares observation i a **high leverage point** when $h_{ii} > 3(k + 1)/n$, three times the mean; ISLR when h_{ii} greatly exceeds $(k + 1)/n$. Leverage depends on $\mathbf{X}$ alone, so replacing the response by a transformation of it leaves every leverage unchanged.

Remark (Options for handling high leverage points; FREES §5.4.1)**.** FREES's options: keep the observation and comment on its effect; delete it as unrepresentative, or present the analysis with and without it; replace the explanatory variable by another carrying similar information; or transform the variable, for example replacing `DEBEQ` by the indicator of `DEBEQ` > 5, which bounds

the distance of every company in that coordinate. Robust estimation, which down-weights unusual observations, is FREES's intermediate course.

3.2 Cook's distance

Definition 4.13 (Cook's distance; FREES §5.4.2)**.** For $h_{ii} < 1$ and $s > 0$, with $\hat{\mathbf{Y}}_{(i)} = \mathbf{X}\hat{\boldsymbol{\beta}}_{(i)}$ the fitted values of all n observations from the regression without observation i,

$$D_i = \frac{\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)}\|^2}{(k+1)s^2} = \frac{\sum_{j=1}^{n}(\hat{y}_j - \hat{y}_{j(i)})^2}{(k+1)s^2}.$$

Theorem 4.14 (Cook's distance from one fit; FREES §5.4.2, §5.10.2)**.** Under A4, with $h_{ii} < 1$ and $s > 0$,

$$D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})}{(k+1)s^2} = \frac{e_i^2\, h_{ii}}{(k+1)s^2(1-h_{ii})^2} = \frac{r_i^2}{k+1}\frac{h_{ii}}{1-h_{ii}}, \tag{4}$$

and $D_i \leqslant (n-k-1)h_{ii}/\{(k+1)(1-h_{ii})\}$.

Proof. $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)} = \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})$, whose squared length is $(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})$. By Proposition 4.3(iii), $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\, e_i/(1-h_{ii})$, so this equals

$$\frac{e_i^2}{(1-h_{ii})^2}\,\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = \frac{e_i^2\, h_{ii}}{(1-h_{ii})^2}.$$

Substituting $e_i^2 = r_i^2 s^2(1-h_{ii})$ gives the last form, and $r_i^2 \leqslant n-k-1$ (Theorem 4.5(i)) the bound. □

Remark (Reading Cook's distance; FREES §5.4.2)**.** D_i is an outlier measure r_i^2 times a leverage measure $h_{ii}/(1-h_{ii})$, which increases from $1/(n-1)$ at $h_{ii} = 1/n$ to ∞ as $h_{ii} \to 1$. FREES's typical size substitutes $r_i^2 = 1$ and $h_{ii} = (k+1)/n$, giving $1/(n-k-1)$, about $1/n$. FREES also compares D_i with $F_{k+1,n-k-1}$. By Unit 3's theorem **the general linear hypothesis** with $\mathbf{R} = \mathbf{I}_{k+1}$, the statistic for $H_0\colon \boldsymbol{\beta} = \boldsymbol{\beta}_0$ is $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)/\{(k+1)s^2\}$, which has the $F_{k+1,n-k-1}$ law at the true $\boldsymbol{\beta}$ under A1–A5, and D_i is its value at $\boldsymbol{\beta}_0 = \hat{\boldsymbol{\beta}}_{(i)}$: D_i exceeds the $1-\alpha$ quantile iff $\hat{\boldsymbol{\beta}}_{(i)}$ lies outside the $100(1-\alpha)\%$ confidence region $\{\boldsymbol{\beta}_0\colon F(\boldsymbol{\beta}_0) \leqslant F_{k+1,n-k-1,1-\alpha}\}$ for $\boldsymbol{\beta}$. D_i is bounded (Theorem 4.14), so it does not have an F law; the quantile is a yardstick.

Example 4.5 (Outliers and high leverage points, continued; FREES §5.4.2)**.** FREES Table 5.6, with Cook's distance computed from the definition by refitting and from Theorem 4.14:

```
tab56 <- sapply(1:3, function(code) {
  d <- subset(ol, CODES %in% c(0, code)); m <- lm(Y ~ X, data = d); i <- nrow(d)
  fit_without <- model.matrix(m) %*% coef(lm(Y ~ X, data = d[-i, ]))
  c(r = rstandard(m)[[i]], h = hatvalues(m)[[i]],
    D_refit = sum((fitted(m) - fit_without)^2) / (2 * summary(m)$sigma^2),
    D_formula = rstandard(m)[[i]]^2 / 2 * hatvalues(m)[[i]] / (1 -
     ↪  hatvalues(m)[[i]]),
    D_R = cooks.distance(m)[[i]])
})
colnames(tab56) <- c("A", "B", "C"); round(tab56, 4)
```

```
##                A      B       C
## r         4.0030 0.7711 -4.0139
## h         0.0672 0.5497  0.5497
## D_refit   0.5768 0.3629  9.8322
## D_formula 0.5768 0.3629  9.8322
## D_R       0.5768 0.3629  9.8322
```

```r
c(high_leverage_cut = 3 * 2 / 20, typical_D = 1 / 20, F_2_18_95 = qf(0.95, 2, 18))
```

```
## high_leverage_cut         typical_D         F_2_18_95
##             0.300             0.050             3.555
```

A and C are outliers ($|r| > 2$); B and C are high leverage points, both with $h = 0.550 > 3(k+1)/n = 0.3$, equal because they share $x = 9.5$. Cook's distance flags all three against FREES's typical value $1/n = 0.05$, and only C, with $D = 9.83$, exceeds the $F_{2,18}$ quantile 3.555, so only deleting C moves $\hat{\boldsymbol{\beta}}$ outside the 95% confidence region.

Definition 4.15 (DFFITS and DFBETAS (beyond the syllabus))**.** For $0 < h_{ii} < 1$ and $s_{(i)} > 0$,

$$\text{DFFITS}_i = \frac{\hat{y}_i - \hat{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}}, \qquad \text{DFBETAS}_{ij} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{s_{(i)}\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}}, \quad j = 0, \ldots, k,$$

where $\hat{y}_{i(i)} = \mathbf{x}_i'\hat{\boldsymbol{\beta}}_{(i)}$ and $\hat{\beta}_{j(i)}$ is the j-th entry of $\hat{\boldsymbol{\beta}}_{(i)}$.

Proposition 4.16 (DFFITS and DFBETAS from one fit (beyond the syllabus))**.** With an intercept, under A4, $h_{ii} < 1$, $s > 0$ and $s_{(i)} > 0$,

$$\text{DFFITS}_i = t_i \sqrt{\frac{h_{ii}}{1 - h_{ii}}}, \qquad D_i = \frac{s_{(i)}^2}{(k+1)s^2} \text{DFFITS}_i^2,$$

$$\text{DFBETAS}_{ij} = \frac{[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i]_j \, e_i}{(1 - h_{ii})\, s_{(i)} \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}}.$$

Proof. With an intercept $h_{ii} \geqslant 1/n > 0$, so the divisions are defined. By Proposition 4.3(iii), $\hat{y}_i - \hat{y}_{i(i)} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i \, e_i/(1 - h_{ii}) = h_{ii}e_i/(1 - h_{ii})$; dividing by $s_{(i)}\sqrt{h_{ii}}$ gives $e_i\sqrt{h_{ii}}/\{s_{(i)}(1 - h_{ii})\} = t_i\sqrt{h_{ii}/(1 - h_{ii})}$. The j-th entry of $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i e_i/(1 - h_{ii})$ gives DFBETAS. Finally $r_i = t_i s_{(i)}/s$, so Theorem 4.14 gives $D_i = t_i^2 s_{(i)}^2 h_{ii}/\{(k+1)s^2(1 - h_{ii})\}$. □

Example 4.6 (Stock market liquidity, continued: IBM; FREES §5.5.2 and chapter 5 script)**.** The regression of `VOLUME` on `PRICE`, `SHARE` and `VALUE`, whose diagnostics close FREES's liquidity

analysis:

```
mv <- lm(VOLUME ~ PRICE + SHARE + VALUE, data = liq); k <- 3
infl <- cbind(h = hatvalues(mv), t = rstudent(mv), D = cooks.distance(mv))
round(infl[order(-infl[, "D"])[1:4], ], 4)
```

```
##           h      t      D
## 60   0.8756  1.217 2.5954
## 122  0.3371 -3.735 1.5990
## 53   0.0411 -3.272 0.1061
## 45   0.0890 -1.660 0.0663
```

```
c(high_leverage_cut = 3 * (k + 1) / n)
```

```
## high_leverage_cut
##           0.09756
```

```
liq[c(60, 122), c("COMPANY", "VOLUME", "PRICE", "SHARE", "VALUE")]
```

```
##                    COMPANY VOLUME  PRICE SHARE VALUE
## 60   INTNATL BUSINESS MACH  64.57 122.38 612.7 75.44
## 122             EXXON CORP  40.58  44.88 783.1 35.24
```

```
Xt <- as.matrix(liq[c("PRICE", "SHARE", "VALUE")]); S <- cov(Xt)
```

```
round(c((Xt[60, ] - colMeans(Xt)) / sqrt(diag(S)), Delta2 = mahalanobis(Xt[60, ],
 ↪  colMeans(Xt), S),
       h_from_Delta2 = 1 / n + mahalanobis(Xt[60, ], colMeans(Xt), S) / (n - 1),
       h_over_1_minus_h = infl[60, "h"] / (1 - infl[60, "h"])), 4)
```

```
##          PRICE          SHARE          VALUE
##         3.9111         4.4999         8.7434
##         Delta2  h_from_Delta2 h_over_1_minus_h
##       105.8308         0.8756         7.0383
```

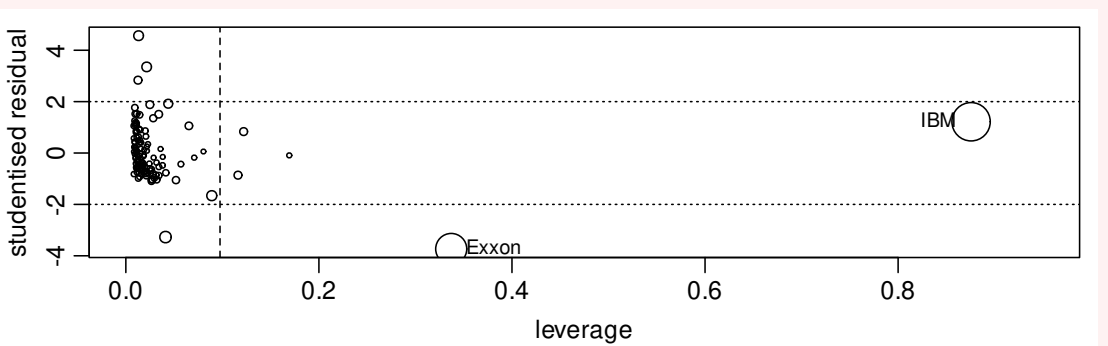


Figure 4.5: Studentised residuals against leverage, circle areas growing with Cook's distance; dashed line at $3(k+1)/n$, dotted lines at ± 2.

IBM has leverage 0.876, nine times the cutoff $3(k+1)/n = 0.098$: its price, shares and value are 3.9, 4.5 and 8.7 standard deviations above the means, and its squared Mahalanobis distance 105.8 reproduces h through (3). Its studentised residual is only 1.22, yet its Cook's distance 2.60 is the largest, through the leverage factor $h/(1-h) = 7.0$. Exxon combines leverage 0.337 with $t = -3.73$. Refitting without IBM:

```
mv60 <- update(mv, data = liq[-60, ])
round(rbind(all = coef(summary(mv))[, 3], without_IBM = coef(summary(mv60))[, 3]),
 ↪  3)
```

```
##             (Intercept)  PRICE SHARE  VALUE
## all               4.939 -0.635 5.194  1.936
## without_IBM       3.831  0.067 4.026 -0.317
```

```
round(rbind(all = coef(mv), without_IBM = coef(mv60)), 5)
```

```
##             (Intercept)    PRICE   SHARE   VALUE
## all               7.909 -0.02224 0.05372  0.3127
## without_IBM       6.899  0.00270 0.07120 -0.1250
```

```
round(rbind(all = car::vif(mv), without_IBM = car::vif(mv60)), 3)
```

```
##             PRICE SHARE  VALUE
## all         1.513 3.827  4.686
## without_IBM 1.775 9.355 10.317
```

```
c(R2_all = summary(mv)$r.squared, R2_without_IBM = summary(mv60)$r.squared)
```

```
##         R2_all R2_without_IBM
##         0.6101         0.5239
```

The coefficient of **VALUE** falls from 0.313 ($t = 1.94$) to -0.125 ($t = -0.32$) and R^2 from 0.610 to 0.524: one company in 123 carries the evidence for **VALUE**.

4 Collinearity

FREES §5.5, §5.10.3 ·ISLR §3.3.3 (collinearity) ·CLM §6. SRM learning outcomes 2(d), 2(e).

4.1 What collinearity does

Example 4.7 (Perfectly correlated explanatory variables; FREES §5.5.1)**.** Joe fits $\hat{y} = -87 + x_1 + 18x_2$ and Jane fits $\hat{y} = -7 + 9x_1 + 2x_2$ to four observations with $x_2 = 5 + x_1/2$:

```
jf <- data.frame(y = c(23, 83, 63, 103), x1 = c(2, 8, 6, 10), x2 = c(6, 9, 8, 10))
Xj <- cbind(1, jf$x1, jf$x2)
cbind(y = jf$y, joe = drop(Xj %*% c(-87, 1, 18)), jane = drop(Xj %*% c(-7, 9, 2)),
      null_direction = drop(Xj %*% c(10, 1, -2)))
```

```
##         y joe jane null_direction
## [1,]   23  23   23              0
## [2,]   83  83   83              0
## [3,]   63  63   63              0
## [4,] 103 103  103              0
```

```
c(rank = qr(Xj)$rank); coef(lm(y ~ x1 + x2, data = jf))
```

```
## rank
##    2
## (Intercept)          x1          x2
##           3          10          NA
```

Both fit every observation exactly, and so does R's $(3, 10, 0)$, which drops x_2.

Proposition 4.17 (Exact collinearity; FREES §5.5.1)**.** If $\operatorname{rank}(\mathbf{X}) < k+1$, the solutions of the normal equations $\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y}$ form the set $\hat{\boldsymbol{\beta}}_0 + \mathcal{N}(\mathbf{X})$, for any one solution $\hat{\boldsymbol{\beta}}_0$, which is infinite; all of them give the same fitted values, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$, hence the same residuals, SSE and R^2.

Proof. By the prerequisites' theorem **orthogonal projection** there is a unique $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ with $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$; writing $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}_0$, this says $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_0) = \mathbf{0}$, so $\hat{\boldsymbol{\beta}}_0$ solves the normal equations. Any solution $\boldsymbol{\beta}$ has $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} \perp \mathcal{C}(\mathbf{X})$, so $\mathbf{X}\boldsymbol{\beta} = \hat{\mathbf{Y}}$ by uniqueness and $\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_0 \in \mathcal{N}(\mathbf{X})$; conversely every $\hat{\boldsymbol{\beta}}_0 + \mathbf{v}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$ solves them. By the prerequisites' rank–nullity theorem, $\dim \mathcal{N}(\mathbf{X}) = k + 1 - \operatorname{rank}(\mathbf{X}) \geqslant 1$. □

Remark (FREES §5.5.1)**.** In Example 4.7, $\mathcal{N}(\mathbf{X})$ is spanned by $(10, 1, -2)'$, since $10+x_1-2x_2 = 0$ on the data; Jane's coefficients are Joe's plus $8(10, 1, -2)'$ and R's are Joe's plus $9(10, 1, -2)'$. The fit is determined and the coefficients are not.

Proposition 4.18 (What collinearity does and does not affect; FREES §5.5.1–5.5.2)**.** Suppose the model has an intercept and A1–A4, and let R_j^2 and s_{x_j} be as in Unit 3's corollary **the standard error through the other explanatory variables**.

(i) $\mathsf{E}[s^2] = \sigma^2$ and $\mathrm{var}(\hat{y}_i) = \sigma^2 h_{ii} \leqslant \sigma^2$ for every i, whatever the correlations among the explanatory variables.

(ii) $\mathrm{var}(\hat{\beta}_j) = \sigma^2 \mathrm{VIF}_j / \{(n-1)s_{x_j}^2\}$, which for fixed s_{x_j} grows without bound as $R_j^2 \to 1$.

(iii) The slope block of $(\mathbf{X}'\mathbf{X})^{-1}$ is $(\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c)^{-1}$; for $k = 2$, $\mathrm{cor}(\hat{\beta}_1, \hat{\beta}_2) = -r_{12}$, the negative of the sample correlation of x_1 and x_2.

Proof. (i) $\mathsf{E}[s^2] = \sigma^2$ is Unit 2's theorem **unbiasedness of the mean square error**. $\hat{\mathbf{Y}} = \mathbf{HY}$, so $\mathrm{var}(\hat{\mathbf{Y}}) = \mathbf{H}(\sigma^2\mathbf{I})\mathbf{H} = \sigma^2\mathbf{H}$, whose diagonal entries are $\sigma^2 h_{ii} \leqslant \sigma^2$ by Unit 2's theorem **properties of leverage**.

(ii) Unit 3's corollary cited in the statement.

(iii) In the proof of Proposition 4.12, $(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{A}\,\mathrm{diag}(1/n, \mathbf{C})\mathbf{A}'$ with $\mathbf{C} = (\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c)^{-1}$ and $\mathbf{A} = \begin{pmatrix} 1 & -\bar{\mathbf{x}}' \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}$; multiplying out,

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1/n + \bar{\mathbf{x}}'\mathbf{C}\bar{\mathbf{x}} & -\bar{\mathbf{x}}'\mathbf{C} \\ -\mathbf{C}\bar{\mathbf{x}} & \mathbf{C} \end{pmatrix}.$$

For $k = 2$, with s_1, s_2 the standard deviations of x_1, x_2 and $r = r_{12}$,

$$\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c = (n-1)\begin{pmatrix} s_1^2 & r s_1 s_2 \\ r s_1 s_2 & s_2^2 \end{pmatrix}, \qquad \mathbf{C} = \frac{1}{(n-1)s_1^2 s_2^2 (1-r^2)}\begin{pmatrix} s_2^2 & -r s_1 s_2 \\ -r s_1 s_2 & s_1^2 \end{pmatrix},$$

$|r| < 1$ by A4. By Unit 2's theorem **variance–covariance matrix**, $\mathrm{cor}(\hat{\beta}_1, \hat{\beta}_2) = C_{12}/\sqrt{C_{11}C_{22}} = -r s_1 s_2/(s_1 s_2) = -r$. □

Remark (Collinearity facts; FREES §5.5.1, ISLR §3.3.3)**.** Part (i) is FREES's first two facts: collinearity prevents neither a good fit nor reliable estimates of σ^2 and of the overall fit. Predictions are another matter: by Proposition 4.12 the variance of the fitted mean at a new $\mathbf{x}_0$ is $\sigma^2\{1/n + \Delta_0^2/(n-1)\}$, small where $\mathbf{x}_0$ follows the pattern of the data and large where it breaks a near-linear relation, even with each coordinate inside its observed range. Part (ii) is the third fact and ISLR's point: large standard errors, small t ratios, and tests of $H_0\colon \beta_j = 0$ that lose power.

Proposition 4.19 (The residual sum of squares around its minimum; ISLR §3.3.3)**.** Suppose the model has an intercept and A4, let $\hat{\boldsymbol{\beta}}_s$ be the least squares slopes and $\tilde{\mathbf{X}}, \tilde{\mathbf{X}}_c$ as in Proposition 4.12. For every $\boldsymbol{\beta}_s \in \mathbb{R}^k$,

$$\min_{\beta_0 \in \mathbb{R}} \|\mathbf{Y} - \beta_0 \mathbf{1} - \tilde{\mathbf{X}}\boldsymbol{\beta}_s\|^2 = \mathrm{SSE} + (\boldsymbol{\beta}_s - \hat{\boldsymbol{\beta}}_s)'\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c(\boldsymbol{\beta}_s - \hat{\boldsymbol{\beta}}_s).$$

The contours in $\boldsymbol{\beta}_s$ are therefore ellipsoids centred at $\hat{\boldsymbol{\beta}}_s$, with axes along the eigenvectors of $\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c$ and semi-axes proportional to the reciprocal square roots of its eigenvalues. For $k = 2$ variables with unit standard deviations and correlation r_{12}, the eigenvalues are $(n-1)(1 \pm r_{12})$ and the longest axis is $\sqrt{(1+|r_{12}|)/(1-|r_{12}|)}$ times the shortest.

Proof. Put $\alpha = \beta_0 + \bar{\mathbf{x}}'\boldsymbol{\beta}_s$, so that $\beta_0\mathbf{1} + \tilde{\mathbf{X}}\boldsymbol{\beta}_s = \alpha\mathbf{1} + \tilde{\mathbf{X}}_c\boldsymbol{\beta}_s$, a bijection $(\beta_0, \boldsymbol{\beta}_s) \leftrightarrow (\alpha, \boldsymbol{\beta}_s)$ that keeps the slopes. By Unit 3's proposition **codings are reparametrisations**, the least squares fit in the design $[\mathbf{1}\ \tilde{\mathbf{X}}_c] = \mathbf{X}\mathbf{A}$ has the same fitted values and coefficients $\mathbf{A}^{-1}\hat{\boldsymbol{\beta}} = (\hat{\alpha}, \hat{\boldsymbol{\beta}}_s')'$ with $\hat{\alpha} = \hat{\beta}_0 + \bar{\mathbf{x}}'\hat{\boldsymbol{\beta}}_s$, so $\mathbf{Y} = \hat{\alpha}\mathbf{1} + \tilde{\mathbf{X}}_c\hat{\boldsymbol{\beta}}_s + \mathbf{e}$ and

$$\mathbf{Y} - \alpha\mathbf{1} - \tilde{\mathbf{X}}_c\boldsymbol{\beta}_s = \mathbf{e} + (\hat{\alpha} - \alpha)\mathbf{1} + \tilde{\mathbf{X}}_c(\hat{\boldsymbol{\beta}}_s - \boldsymbol{\beta}_s).$$

The three terms are mutually orthogonal: $\mathbf{e} \perp \mathcal{C}(\mathbf{X})$, and $\mathbf{1}'\tilde{\mathbf{X}}_c = \mathbf{0}'$. By the prerequisites' Pythagoras theorem the squared length is $\mathrm{SSE} + n(\hat{\alpha} - \alpha)^2 + (\hat{\boldsymbol{\beta}}_s - \boldsymbol{\beta}_s)'\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c(\hat{\boldsymbol{\beta}}_s - \boldsymbol{\beta}_s)$, minimised over α at $\alpha = \hat{\alpha}$. By the prerequisites' spectral theorem, $\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal; in the coordinates $\mathbf{w} = \mathbf{P}'(\boldsymbol{\beta}_s - \hat{\boldsymbol{\beta}}_s)$ the contour at level $\mathrm{SSE} + c$ is $\sum_j \lambda_j w_j^2 = c$, with semi-axes $\sqrt{c/\lambda_j}$ along the columns of $\mathbf{P}$. For $k = 2$ unit-variance variables, $\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c = (n-1)\begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}$, with eigenvectors $(1, 1)'/\sqrt{2}$ and $(1, -1)'/\sqrt{2}$ and eigenvalues $(n-1)(1 \pm r_{12})$; the axes are in the ratio $\sqrt{(1 + |r_{12}|)/(1 - |r_{12}|)}$. □

Example 4.8 (Credit; ISLR §3.3.3). ISLR Table 3.11: `Balance` on `Age` and `Limit`, and on `Rating` and `Limit`.

```
cm1 <- lm(Balance ~ Age + Limit, data = Credit); cm2 <- lm(Balance ~ Rating + Limit,
 ↪  data = Credit)
round(coef(summary(cm1)), 4); round(coef(summary(cm2)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -173.4109    43.8284  -3.957   0.0001
## Age           -2.2915     0.6725  -3.408   0.0007
## Limit          0.1734     0.0050  34.496   0.0000
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -377.5368    45.2542  -8.343   0.0000
## Rating         2.2017     0.9523   2.312   0.0213
## Limit          0.0245     0.0638   0.384   0.7012
```

```
c(se_ratio_Limit = coef(summary(cm2))["Limit", 2] / coef(summary(cm1))["Limit", 2],
  r_Age_Limit = cor(Credit$Age, Credit$Limit), r_Rating_Limit = cor(Credit$Rating,
   ↪  Credit$Limit))
```

```
## se_ratio_Limit    r_Age_Limit r_Rating_Limit
##        12.7017         0.1009          0.9969
```

```r
axis_ratio <- function(r) sqrt((1 + abs(r)) / (1 - abs(r)))
c(Age_Limit = axis_ratio(cor(Credit$Age, Credit$Limit)),
  Rating_Limit = axis_ratio(cor(Credit$Rating, Credit$Limit)))
```

```
##    Age_Limit Rating_Limit
##        1.107       25.298
```

```r
V2 <- vcov(cm2); c(cor_b_Rating_b_Limit = V2[2, 3] / sqrt(V2[2, 2] * V2[3, 3]))
```

```
## cor_b_Rating_b_Limit
##              -0.9969
```

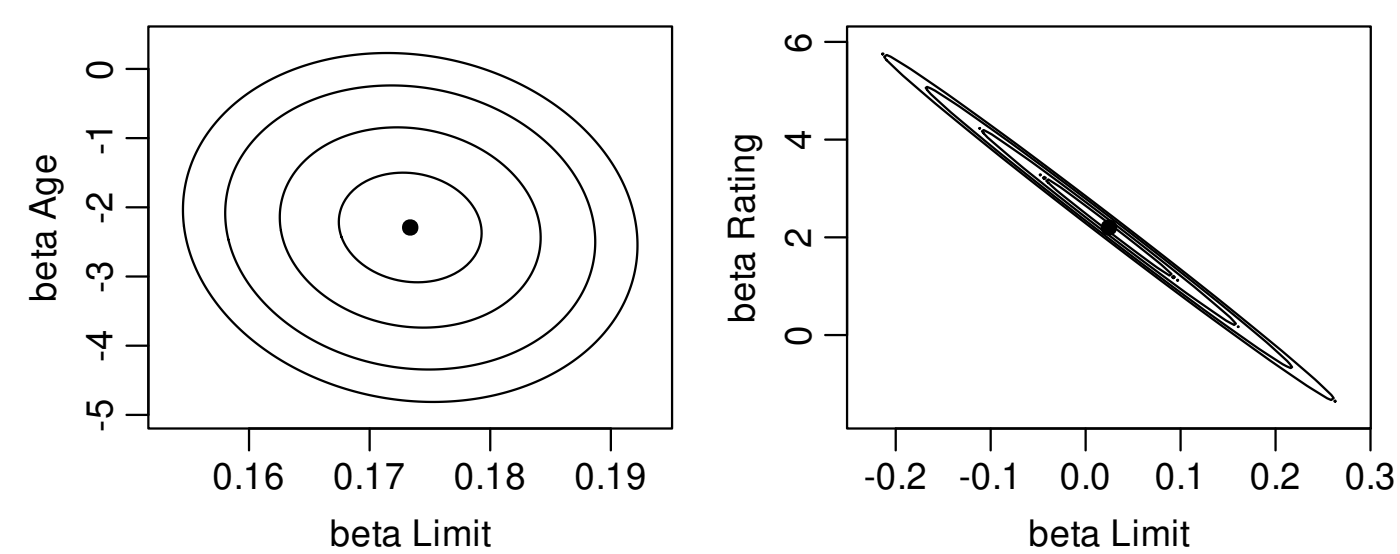


Figure 4.6: Contours of the residual sum of squares, the intercept minimised out, over ±4 standard errors of the two slopes, at $\text{SSE} + 2s^2 F_{2,n-3,1-\alpha}$ for $\alpha = 0.5, 0.1, 0.01, 0.001$ (ISLR Figure 3.15).

`Limit` and `Rating` have correlation 0.997, and in their regression the standard error of the `Limit` coefficient is 12.7 times that in the regression with `Age`, so `Limit` loses its significance ($p = 0.70$).

By Proposition 4.19 the standardised ellipses have axis ratio 25.3 against 1.11 with `Age`: the long valley of the right panel, along which the two coefficients trade off with correlation -0.997 (Proposition 4.18(iii)). The contours drawn are the boundaries of the joint confidence regions for the two slopes: their quadratic form $(\boldsymbol{\beta}_s - \hat{\boldsymbol{\beta}}_s)'\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c(\boldsymbol{\beta}_s - \hat{\boldsymbol{\beta}}_s)$ is, by Proposition 4.18(iii), the numerator of Unit 3's general linear hypothesis statistic for the two slopes, with $q = 2$.

4.2 Variance inflation factors

Proposition 4.20 (Variance inflation factors; FREES §5.5.2, ISLR §3.3.3)**.** Suppose the model has an intercept, $k \geqslant 2$ and A4; part (ii) also assumes A1–A3.

(i) $\mathrm{VIF}_j \geqslant 1$, with equality iff x_j has sample correlation 0 with every other explanatory variable.

(ii) VIF_j is the variance of $\hat{\beta}_j$ in the full regression divided by the variance of the slope of the simple regression of y on x_j alone, ISLR's definition.

(iii) The **tolerance** $1/\mathrm{VIF}_j$ equals $1 - R_j^2$; $\mathrm{VIF}_j > 10$ iff $R_j^2 > 0.9$, and $\mathrm{VIF}_j > 5$ iff $R_j^2 > 0.8$.

(iv) Centring and rescaling the explanatory variables leave every VIF_j unchanged.

Proof. **(i)** Let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ without $\mathbf{x}_j$ and $\mathbf{Z}_c$ its $k-1$ non-constant columns centred. By (2) for the design $\mathbf{X}_{(j)}$, the regression of $\mathbf{x}_j$ on the other variables has regression sum of squares $\|\mathbf{Z}_c(\mathbf{Z}_c'\mathbf{Z}_c)^{-1}\mathbf{Z}_c'\mathbf{x}_j\|^2 \geqslant 0$, so $R_j^2 \geqslant 0$, and $R_j^2 < 1$ by Unit 3's corollary **the standard error through the other explanatory variables**. Hence $\mathrm{VIF}_j = 1/(1-R_j^2) \geqslant 1$, with equality iff $\mathbf{Z}_c(\mathbf{Z}_c'\mathbf{Z}_c)^{-1}\mathbf{Z}_c'\mathbf{x}_j = \mathbf{0}$, which, multiplying by $\mathbf{Z}_c'$, holds iff $\mathbf{Z}_c'\mathbf{x}_j = \mathbf{0}$. The entries of $\mathbf{Z}_c'\mathbf{x}_j$ are $(n-1)$ times the sample covariances of x_j with the other variables, $\mathbf{Z}_c$ having columns that sum to zero.
(ii) The slope of the simple regression is $\sum_i w_i y_i$ with $\sum_i w_i^2 = 1/\{(n-1)s_{x_j}^2\}$ (Unit 2's proposition **coefficients are weighted sums of the responses**), so under A3 its variance is $\sigma^2/\{(n-1)s_{x_j}^2\}$ by the prerequisites' proposition **variance under a linear map**, whether or not its mean is correct; divide Unit 3's formula $\mathrm{var}(\hat{\beta}_j) = \sigma^2 \mathrm{VIF}_j/\{(n-1)s_{x_j}^2\}$ by it.
(iii) $1/(1-R_j^2) > 10$ iff $1 - R_j^2 < 0.1$; likewise for 5.
(iv) Replacing each x_l by $a_l x_l + c_l$ with $a_l \neq 0$ replaces $\mathbf{X}_{(j)}$ by $\mathbf{X}_{(j)}\mathbf{B}$ for an invertible $\mathbf{B}$, which leaves its column space and hat matrix unchanged, and replaces $\mathbf{x}_j$ by $a_j\mathbf{x}_j + c_j\mathbf{1}$. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, the new residuals are a_j times the old and the new total sum of squares is a_j^2 times the old, so $R_j^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ is unchanged. □

Example 4.9 (Credit and liquidity; ISLR §3.3.3, FREES §5.5.2)**.** ISLR's three-variable `Credit` regression and FREES Table 5.7, with FREES's equation (5.5) checked for each coefficient:

```
cm3 <- lm(Balance ~ Age + Rating + Limit, data = Credit)
round(car::vif(cm3), 2)
```

```
##    Age Rating  Limit
##   1.01 160.67 160.59
```

```
c(R2_all_three = summary(cm3)$r.squared, R2_Age_Limit = summary(cm1)$r.squared)
```

```
## R2_all_three R2_Age_Limit
##       0.7536       0.7498
```

```
cat(sprintf("cor(PRICE*SHARE/1000, VALUE) = %.7f, max gap = %.2f\n",
    cor(liq$PRICE * liq$SHARE / 1000, liq$VALUE), max(abs(liq$PRICE * liq$SHARE /
     ↪  1000 - liq$VALUE))))
```

```
## cor(PRICE*SHARE/1000, VALUE) = 0.9999955, max gap = 0.46
```

```
round(cbind(s_x = sapply(liq[c("PRICE", "SHARE", "VALUE")], sd),
 ↪  coef(summary(mv))[-1, 1:3],
            VIF = car::vif(mv),
```

```
            se_from_VIF = summary(mv)$sigma * sqrt(car::vif(mv)) /
              (sapply(liq[c("PRICE", "SHARE", "VALUE")], sd) * sqrt(n - 1))), 4)
```

```
##           s_x Estimate Std. Error t value   VIF se_from_VIF
## PRICE  21.370  -0.0222     0.0350 -0.6349 1.512      0.0350
## SHARE 115.103   0.0537     0.0103  5.1935 3.827      0.0103
## VALUE   8.157   0.3127     0.1615  1.9363 4.686      0.1615
```

```
c(R2 = summary(mv)$r.squared, s = summary(mv)$sigma,
  VIF_PRICE_aux = 1 / (1 - summary(lm(PRICE ~ SHARE + VALUE, data =
   ↪  liq))$r.squared))
```

```
##            R2             s VIF_PRICE_aux
##        0.6101        6.7221        1.5125
```

The **Credit** factors are 1.01, 160.67 and 160.59: **Rating** and **Limit** each have $R_j^2 > 0.99$ on the others. For the liquidity regression, $R^2 = 61\%$, $s = 6.72$, and the factors 1.5, 3.8 and 4.7 are all below 10, although **VALUE** is **PRICE** $\times$ **SHARE**/1000 to within 0.46 billion (correlation 0.999996): the relation is multiplicative, not linear, and the variance inflation factor measures only linear dependence.

Proposition 4.21 (Centring a variable and its square; FREES §5.5.2)**.** Let $x_1, \dots, x_n$ take at least three distinct values and let $c_i = x_i - \bar{x}$. The regressions of y on (x, x^2) and on (c, c^2) have the same fitted values, and the sample correlation of c and c^2 is 0 iff $\sum_i c_i^3 = 0$, which holds when the x_i are symmetric about $\bar{x}$.

Proof. $c = x - \bar{x}\mathbf{1}$ and $c^2 = x^2 - 2\bar{x}\,x + \bar{x}^2\mathbf{1}$ entrywise, so $[\mathbf{1}\ c\ c^2] = [\mathbf{1}\ x\ x^2]\mathbf{A}$ with $\mathbf{A}$ upper triangular with unit diagonal; three distinct values make $[\mathbf{1}\ x\ x^2]$ of full rank (a non-zero quadratic has at most two roots), and Unit 3's proposition **codings are reparametrisations** gives the same fitted values. If the c_i^2 were all equal to some d, every c_i would be $\sqrt{d}$ or $-\sqrt{d}$ and the x_i would take at most two values; so the c_i^2 are not all equal. With $\sum_i c_i = 0$, the sample covariance of c and c^2 is $\sum_i c_i(c_i^2 - \overline{c^2})/(n-1) = \sum_i c_i^3/(n-1)$. Symmetry about $\bar{x}$ pairs each c_i with $-c_i$ and makes $\sum_i c_i^3 = 0$. □

Example 4.10 (Centring horsepower; FREES §5.5.2)**.**

```
hp <- Auto$horsepower; hc <- hp - mean(hp)
q_raw <- lm(Auto$mpg ~ hp + I(hp^2)); q_cen <- lm(Auto$mpg ~ hc + I(hc^2))
```

```r
range(hp)
```

```
## [1]  46 230
```

```r
round(c(cor_raw = cor(hp, hp^2), VIF_raw = 1 / (1 - cor(hp, hp^2)^2),
  cor_centred = cor(hc, hc^2), VIF_centred = 1 / (1 - cor(hc, hc^2)^2)), 3)
```

```
##     cor_raw     VIF_raw cor_centred VIF_centred
##       0.983      29.334       0.663       1.782
```

```r
signif(c(sum_c_cubed_over_n = mean(hc^3), fitted_diff = max(abs(fitted(q_raw) -
 ↪  fitted(q_cen)))), 3)
```

```
## sum_c_cubed_over_n        fitted_diff
##           6.15e+04           1.31e-13
```

Over the sampled range, 46 to 230, **horsepower** and its square are nearly linearly related, with correlation 0.983 and variance inflation factor 29.3 (FREES Figure 5.5). Centring lowers the correlation to 0.663 and the factor to 1.78 without changing the fit; not to 0, because the distribution of **horsepower** is skewed to the right and $\sum_i c_i^3 > 0$.

Remark (Remedies; FREES §5.5.2, ISLR §3.3.3)**.** Rescaling variables of very different magnitudes is a computational remedy only: by Proposition 4.20(iv) it changes no variance inflation factor, and by Unit 3's proposition **codings are reparametrisations** no fit. The statistical remedies are to drop one of the collinear variables or to combine them. In Example 4.9, dropping `Rating` lowers R^2 from 0.7536 to 0.7498.

```
Credit$worth <- drop(scale(Credit$Limit) + scale(Credit$Rating)) / 2
cm4 <- lm(Balance ~ Age + worth, data = Credit)
c(R2 = summary(cm4)$r.squared, car::vif(cm4))
```

```
##     R2    Age  worth
## 0.7529 1.0105 1.0105
```

The average of the standardised `Limit` and `Rating`, a creditworthiness score, keeps $R^2 = 0.7529$ with both variance inflation factors at 1.01.

4.3 Collinearity and leverage

Proposition 4.22 (One point and collinearity; FREES §5.5.3)**.** Let $(u_1, v_1), \dots, (u_m, v_m)$ be fixed, $m \geqslant 2$, add the point (ca, cb) with $a \neq 0 \neq b$, and let $r(c)$ be the sample correlation of the $m+1$ points.

(i) $r(c) \to \operatorname{sign}(ab)$ as $c \to \infty$, so the variance inflation factor $1/\{1 - r(c)^2\}$ of either variable in a regression on both grows without bound.

(ii) If the m points are not all equal and lie on one line, and (ca, cb) is off that line, then $|r(c)| < 1$.

Proof. **(i)** With $N = m+1$, $U = \sum_{i \leqslant m} u_i$, $V = \sum_{i \leqslant m} v_i$ and $P = \sum_{i \leqslant m} u_i v_i$, the centred cross product of the N points is

$$S_{uv}(c) = P + c^2 ab - \frac{(U + ca)(V + cb)}{N} = c^2 ab\Big(1 - \frac{1}{N}\Big) - c\,\frac{aV + bU}{N} + P - \frac{UV}{N},$$

and likewise $S_{uu}(c) = c^2 a^2 (1 - 1/N) + O(c)$, $S_{vv}(c) = c^2 b^2 (1 - 1/N) + O(c)$. Dividing numerator and denominator of $r(c) = S_{uv}/\sqrt{S_{uu} S_{vv}}$ by c^2, $r(c) \to ab/\sqrt{a^2 b^2} = \operatorname{sign}(ab)$. For $k = 2$,

$R_1^2 = R_2^2 = r^2$, Unit 2's identity $R^2 = r^2$ for one explanatory variable.
(ii) By the prerequisites' Cauchy–Schwarz theorem, $|r| = 1$ iff the centred vectors $(u_i - \bar{u})$ and $(v_i - \bar{v})$ are proportional, that is iff all N points lie on one line. The only line through the m points, which are not all equal, is the given one, and (ca, cb) is off it. □

Example 4.11 (Masking and inducing; FREES §5.5.3)**.** Thirty nearly collinear points plus one point off their line, and thirty uncorrelated points plus one point moved out along the diagonal (FREES Figures 5.6 and 5.7):

```
set.seed(11); x1 <- rnorm(30); x2 <- x1 + rnorm(30, sd = 0.15)
vif2 <- function(a, b) 1 / (1 - cor(a, b)^2)
c(r_30 = cor(x1, x2), VIF_30 = vif2(x1, x2),
  r_31 = cor(c(x1, 2.5), c(x2, -2.5)), VIF_31 = vif2(c(x1, 2.5), c(x2, -2.5)))
```

```
##    r_30  VIF_30    r_31  VIF_31
##  0.9825 28.7616  0.4327  1.2303
```

```
z1 <- rnorm(30); z2 <- rnorm(30)
round(sapply(c(0, 5, 20, 100), function(cc)
  c(c = cc, r = cor(c(z1, cc), c(z2, cc)), VIF = vif2(c(z1, cc), c(z2, cc)))), 3)
```

```
##       [,1]  [,2]   [,3]    [,4]
## c    0.000 5.000 20.000 100.000
## r   -0.182 0.349  0.916   0.996
## VIF  1.034 1.139  6.194 138.321
```

One point off the line of the first thirty cuts the variance inflation factor from 28.8 to 1.23; moving one point out along the diagonal of the second cloud drives the correlation from -0.18 to 0.996 at $c = 100$, with variance inflation factor 138, as Proposition 4.22(i) predicts. The same happens in Example 4.6: without IBM the factors of `SHARE` and `VALUE` rise from 3.8 and 4.7 to 9.4 and 10.3, so the high leverage point was masking the collinearity, and through Unit 3's standard error formula it moved the standard errors too.

4.4 Suppressor variables

Proposition 4.23 (R^2 from correlations; FREES §5.5.4)**.** For the regression of a non-constant y on $x_1, \ldots, x_k$ with an intercept, let $\mathbf{r}_y$ be the vector of sample correlations $r(y, x_j)$ and $\mathbf{R}_{xx}$ the

correlation matrix of the x's, invertible under A4. Then $R^2 = \mathbf{r}_y' \mathbf{R}_{xx}^{-1} \mathbf{r}_y$, and for $k = 2$

$$R^2 = \frac{r_{y1}^2 + r_{y2}^2 - 2 r_{y1} r_{y2} r_{12}}{1 - r_{12}^2}. \tag{5}$$

Proof. By (2), $\text{SSR} = \|(\mathbf{H} - \frac{1}{n}\mathbf{1}\mathbf{1}')\mathbf{Y}\|^2 = \mathbf{Y}'\tilde{\mathbf{X}}_c(\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c)^{-1}\tilde{\mathbf{X}}_c'\mathbf{Y}$. With $\mathbf{D} = \text{diag}(s_{x_1}, \ldots, s_{x_k})$ and s_y the standard deviation of y, $\tilde{\mathbf{X}}_c'\mathbf{Y} = (n-1)s_y \mathbf{D}\mathbf{r}_y$ and $\tilde{\mathbf{X}}_c'\tilde{\mathbf{X}}_c = (n-1)\mathbf{D}\mathbf{R}_{xx}\mathbf{D}$, so $\text{SSR} = (n-1)s_y^2 \mathbf{r}_y' \mathbf{R}_{xx}^{-1} \mathbf{r}_y$; divide by $\text{SST} = (n-1)s_y^2 > 0$. For $k = 2$, $\mathbf{R}_{xx}^{-1} = (1-r_{12}^2)^{-1} \begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix}$. □

Example 4.12 (Suppressor variable; FREES §5.5.4)**.** FREES Table 5.8 reports $r_{12} = 0.972$, $r_{y1} = 0.188$ and $r_{y2} = -0.022$ for 50 hypothetical observations, and $R^2 = 80.7\%$ for the regression on both:

```
R2k2 <- function(r1, r2, r12) (r1^2 + r2^2 - 2 * r1 * r2 * r12) / (1 - r12^2)
g <- expand.grid(r1 = c(0.1875, 0.1885), r2 = c(-0.0225, -0.0215), r12 = c(0.9715,
 ↪  0.9725))
```

```r
c(printed = R2k2(0.188, -0.022, 0.972), min_rounding = min(with(g, R2k2(r1, r2,
  r12))),
  max_rounding = max(with(g, R2k2(r1, r2, r12))), r_y2_zero = R2k2(0.188, 0, 0.972),
  r_y1_sq = 0.188^2, one_minus_r12_sq = 1 - 0.972^2)
```

```
##          printed     min_rounding     max_rounding
##          0.79449          0.77332          0.81646
##       r_y2_zero          r_y1_sq one_minus_r12_sq
##          0.64010          0.03534          0.05522
```

```r
set.seed(5); x1 <- rnorm(50); x2 <- 0.972 * x1 + sqrt(1 - 0.972^2) * rnorm(50)
y <- 3 * (x1 - x2) + 0.3 * rnorm(50)
round(cor(cbind(x1, x2, y)), 3)
```

```
##       x1     x2      y
## x1 1.000  0.984  0.049
## x2 0.984  1.000 -0.100
## y  0.049 -0.100  1.000
```

```r
c(R2_x1 = summary(lm(y ~ x1))$r.squared, R2_x2 = summary(lm(y ~ x2))$r.squared,
  R2_both = summary(lm(y ~ x1 + x2))$r.squared)
```

```
##    R2_x1    R2_x2  R2_both
## 0.002420 0.009931 0.711597
```

The printed correlations give $R^2 = 0.794$ by (5), and correlations that round to them give anything from 0.773 to 0.816, which contains FREES's 0.807: the denominator $1 - r_{12}^2 = 0.055$ magnifies rounding. With $r_{y2} = 0$ the formula is $r_{y1}^2/(1 - r_{12}^2)$, so a variable uncorrelated with y raises R^2 from 0.035 to 0.640; it **suppresses** the part of x_1 unrelated to y. The simulated data reproduce the pattern: R^2 of 0.2% and 1.0% separately and 71% together.

4.5 Orthogonal variables

Proposition 4.24 (Orthogonal explanatory variables; FREES §5.5.5)**.** Let $\mathbf{X} = [\mathbf{X}_1\ \mathbf{X}_2]$ satisfy A4 with $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ and $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$.

(i) The coefficients of $\mathbf{X}_1$ are the same with or without $\mathbf{X}_2$.

(ii) Adding $\mathbf{X}_2$ reduces SSE by $\mathbf{Y}'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{Y}$, which does not involve $\mathbf{X}_1$.

(iii) If $\mathbf{X}_2$ is one column $\mathbf{x}_2$, its variance inflation factor is 1.

Proof. $\mathbf{X}'\mathbf{X} = \operatorname{diag}(\mathbf{X}_1'\mathbf{X}_1, \mathbf{X}_2'\mathbf{X}_2)$, so $\hat{\boldsymbol{\beta}} = \left((\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y},\ (\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{Y}\right)$, whose first block is the coefficient vector of the regression on $\mathbf{X}_1$ alone: (i). Likewise $\mathbf{H} = \mathbf{H}_1 + \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$ with $\mathbf{H}_1$ the hat matrix of $\mathbf{X}_1$, and $\mathrm{SSE}_1 - \mathrm{SSE} = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_1)\mathbf{Y}$: (ii). For (iii), $\mathbf{1}'\mathbf{x}_2 = 0$, so x_2 has mean 0 and its sample covariance with any column $\mathbf{x}_l$ of $\mathbf{X}_1$ is $\mathbf{x}_2'\mathbf{x}_l/(n-1) = 0$; Proposition 4.20(i) applies, and if $\mathbf{X}_1 = \mathbf{1}$ the regression of $\mathbf{x}_2$ on the intercept alone has $R^2 = 0$, as in Unit 3's corollary **the standard error through the other explanatory variables**. □

Example 4.13 (Orthogonal polynomials; FREES §5.5.5). `poly(horsepower, 3)` has orthonormal columns orthogonal to **1**, as the second printed line checks, so Proposition 4.24 applies to the cubic column:

```
p2 <- lm(mpg ~ poly(horsepower, 2), data = Auto); p3 <- lm(mpg ~ poly(horsepower,
 ↪  3), data = Auto)
rbind(degree_2 = coef(p2), degree_3 = coef(p3)[1:3])
```

```
##          (Intercept) poly(horsepower, 2)1
## degree_2       23.45               -120.1
## degree_3       23.45               -120.1
```

```
##          poly(horsepower, 2)2
## degree_2                44.09
## degree_3                44.09
```

```
P3 <- poly(Auto$horsepower, 3)
c(max_offset_from_identity = max(abs(crossprod(P3) - diag(3))), max_col_sum =
  ↪  max(abs(colSums(P3))))
```

```
## max_offset_from_identity             max_col_sum
##                6.661e-16               3.743e-16
```

```
c(VIF = sapply(1:3, function(j) 1 / (1 - summary(lm(P3[, j] ~ P3[,
  ↪  -j]))$r.squared)),
  SSE_drop = sum(residuals(p2)^2) - sum(residuals(p3)^2), cubic_coef_sq =
    ↪  coef(p3)[[4]]^2)
```

```
##          VIF1          VIF2          VIF3      SSE_drop
##          1.00          1.00          1.00         15.59
## cubic_coef_sq
##         15.59
```

The reduction in SSE from the cubic column, 15.59, is the square of its coefficient, as (ii) gives for a

column of unit length. FREES's principal components, a transformation $\mathbf{X}^* = \mathbf{XP}$ to orthogonal columns, are of the same kind: by Unit 3's proposition **codings are reparametrisations**, the fit is unchanged and $\boldsymbol{\beta}^* = \mathbf{P}^{-1}\boldsymbol{\beta}$.

5 Heteroscedasticity

FREES §5.7 ·ISLR §3.3.3 (non-constant variance of error terms). SRM learning outcomes 2(d), 2(e).

Definition 4.25 (Heteroscedasticity; FREES §5.7)**.** The errors are **heteroscedastic** when A3 is replaced by $\text{var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Omega} = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ with the σ_i^2 not all equal: the errors remain uncorrelated and their variances vary by observation. A3 itself is **homoscedasticity**.

Remark (FREES §5.7)**.** Proposition 4.9 with $\boldsymbol{\Sigma} = \boldsymbol{\Omega}$ gives the consequences: $\hat{\boldsymbol{\beta}}$ is still unbiased, its variance is FREES's (5.11),

$$\text{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{n} \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'\Big)(\mathbf{X}'\mathbf{X})^{-1},$$

while the reported $s^2(\mathbf{X}'\mathbf{X})^{-1}$ has expectation $\{\sum_i(1-h_{ii})\sigma_i^2/(n-k-1)\}(\mathbf{X}'\mathbf{X})^{-1}$, since $\operatorname{tr}(\mathbf{M\Omega}) = \sum_i(1-h_{ii})\sigma_i^2$. The Gauss–Markov theorem of Unit 2 assumes A3 and no longer applies.

5.1 Detecting heteroscedasticity

Example 4.14 (A funnel; FREES §5.7.1, ISLR §3.3.3)**.** The quadratic regression of `mpg` on `horsepower` of Example 4.3, and the same regression for $\ln$ `mpg` (ISLR Figure 3.11):

```
l2 <- lm(log(mpg) ~ horsepower + I(horsepower^2), data = Auto)
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.4, 1.4, 0.6), mgp = c(1.9, 0.6, 0))
for (m in list(a2, l2)) {
  plot(fitted(m), residuals(m), xlab = "fitted", ylab = "residual", pch = 16, cex =
   ↪  0.4,
       main = if (identical(m, a2)) "mpg" else "log(mpg)", cex.main = 0.9)
  abline(h = 0, lty = 2)
}
par(op)
```

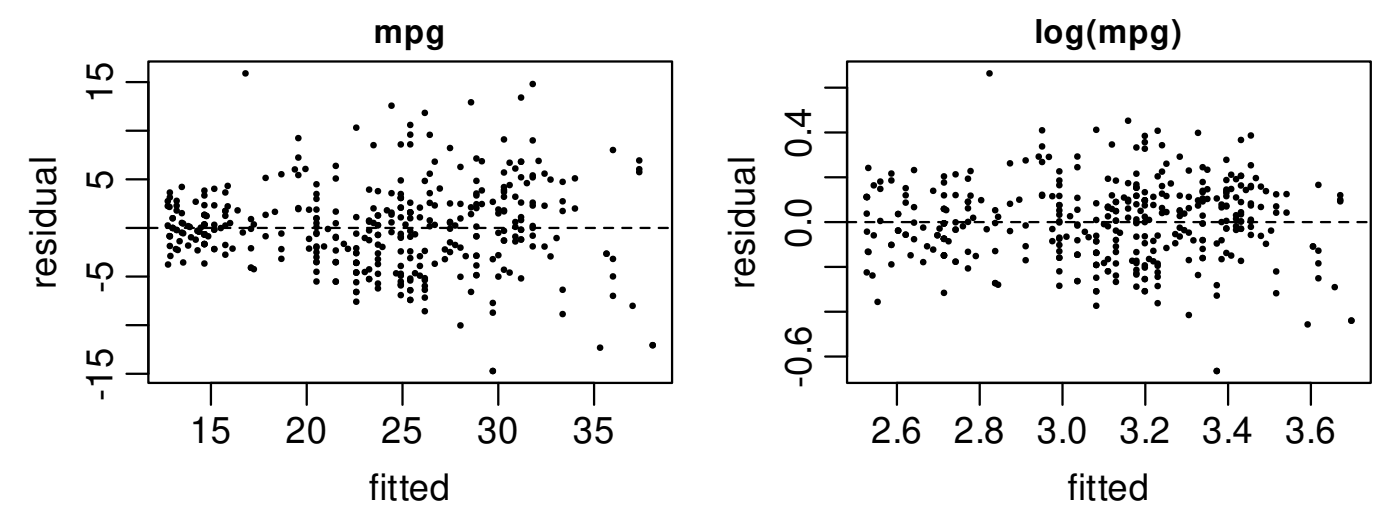


Figure 4.7: Residuals against fitted values for `mpg` and for ln `mpg` on `horsepower` and its square.

For `mpg` the residuals fan out as the fitted value grows, a funnel; for ln `mpg` their spread is nearly the same across the fitted range. Residuals are plotted against fitted values because the fitted value estimates the mean, with which the variance often grows.

Definition 4.26 (The Breusch–Pagan test; FREES §5.7.1)**.** For known vectors $\mathbf{z}_i \in \mathbb{R}^p$, the alternative is H_a: $\operatorname{var}(y_i) = \sigma^2 + \mathbf{z}_i'\boldsymbol{\gamma}$ and the null hypothesis H_0: $\boldsymbol{\gamma} = \mathbf{0}$, which is homoscedasticity. FREES's procedure: (i) fit the regression and compute e_i; (ii) compute $e_i^{*2} = e_i^2/s^2$; (iii) regress e_i^{*2} on $\mathbf{z}_i$ with an intercept; (iv) $\mathrm{LM} = \frac{1}{2}\,\mathrm{SSR}_z$, half the regression sum of squares of step (iii); (v) reject H_0 when LM exceeds the $1-\alpha$ quantile of χ^2_p. The same procedure with $\tilde{\sigma}^2 = \mathrm{SSE}/n$ in place of s^2 in step (ii) is also used.

Lemma 4.27 (Residuals in place of errors)**.** For every n let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of rank $k+1$ and rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$, the first n terms of a fixed sequence with $\|\mathbf{x}_i\| \leqslant B$, and suppose $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X$ positive definite. Let $\varepsilon_1, \varepsilon_2, \ldots$ be uncorrelated with mean 0 and $\operatorname{var}(\varepsilon_i) = \sigma_i^2 \leqslant C$. If $\mathbf{a}_{n1}, \ldots, \mathbf{a}_{nn}$ are non-random vectors in $\mathbb{R}^q$ with $\|\mathbf{a}_{ni}\| \leqslant A$, then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbf{a}_{ni} \left(e_i^2 - \varepsilon_i^2\right) \xrightarrow{p} \mathbf{0}.$$

Proof. Let $\boldsymbol{\delta} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, so $e_i = \varepsilon_i - \mathbf{x}_i'\boldsymbol{\delta}$ and $e_i^2 - \varepsilon_i^2 = (\mathbf{x}_i'\boldsymbol{\delta})^2 - 2\varepsilon_i \mathbf{x}_i'\boldsymbol{\delta}$. The sum is $\mathbf{T}_1 - 2\mathbf{T}_2$ with $\mathbf{T}_1 = n^{-1/2}\sum_i \mathbf{a}_{ni}(\mathbf{x}_i'\boldsymbol{\delta})^2$ and $\mathbf{T}_2 = n^{-1/2}\sum_i \mathbf{a}_{ni}\varepsilon_i \mathbf{x}_i'\boldsymbol{\delta}$.

Step 1: $\mathbf{T}_1$. $\|\mathbf{T}_1\| \leqslant A n^{-1/2}\sum_i (\mathbf{x}_i'\boldsymbol{\delta})^2 = A n^{-1/2}\|\mathbf{X}\boldsymbol{\delta}\|^2 = A n^{-1/2}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$, since $\mathbf{X}\boldsymbol{\delta} = \mathbf{H}\boldsymbol{\varepsilon}$. By the prerequisites' lemma **expectation of a quadratic form**, $\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}] = \sum_i h_{ii}\sigma_i^2 \leqslant C\,\mathrm{tr}(\mathbf{H}) = C(k+1)$. So $\mathsf{E}\,\|\mathbf{T}_1\| \leqslant AC(k+1)/\sqrt{n} \to 0$, and $\mathbf{T}_1 \xrightarrow{p} \mathbf{0}$ by Markov's inequality.

Step 2: $\mathbf{T}_2$. $\mathbf{T}_2 = \mathbf{G}_n\boldsymbol{\delta}$ with $\mathbf{G}_n = n^{-1/2}\sum_i \varepsilon_i \mathbf{a}_{ni}\mathbf{x}_i'$, a $q \times (k+1)$ random matrix. By Cauchy–Schwarz on each row, $\|\mathbf{G}_n\boldsymbol{\delta}\| \leqslant \|\mathbf{G}_n\|_F\|\boldsymbol{\delta}\|$, where $\|\cdot\|_F$ is the root sum of squares of the entries, and $2ab \leqslant a^2 + b^2$ gives $\|\mathbf{G}_n\|_F\|\boldsymbol{\delta}\| \leqslant \frac{1}{2}(n^{-1/2}\|\mathbf{G}_n\|_F^2 + n^{1/2}\|\boldsymbol{\delta}\|^2)$. The errors are uncorrelated with mean 0, so the cross terms vanish and $\mathsf{E}\,\|\mathbf{G}_n\|_F^2 = n^{-1}\sum_i \sigma_i^2\|\mathbf{a}_{ni}\|^2\|\mathbf{x}_i\|^2 \leqslant CA^2B^2$. By the quadratic-form lemma again and the cyclic property of the trace,

$$\mathsf{E}\,\|\boldsymbol{\delta}\|^2 = \sum_i \sigma_i^2\,\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-2}\mathbf{x}_i \leqslant C\,\mathrm{tr}\big(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'\big) = C\,\mathrm{tr}\big((\mathbf{X}'\mathbf{X})^{-1}\big) = \frac{C}{n}\,\mathrm{tr}\big((n^{-1}\mathbf{X}'\mathbf{X})^{-1}\big).$$

By the prerequisites' lemma **inverses of convergent positive definite matrices**, the entries of $(n^{-1}\mathbf{X}'\mathbf{X})^{-1}$ are bounded for all large n, so $\mathsf{E}\,\|\mathbf{T}_2\| \leqslant \frac{1}{2}n^{-1/2}\{CA^2B^2 + C\,\mathrm{tr}((n^{-1}\mathbf{X}'\mathbf{X})^{-1})\} \to 0$. Markov's inequality gives $\mathbf{T}_2 \xrightarrow{p} \mathbf{0}$. Finally $\mathsf{P}\{\|\mathbf{T}_1 - 2\mathbf{T}_2\| > \eta\} \leqslant \mathsf{P}\{\|\mathbf{T}_1\| > \eta/2\} + \mathsf{P}\{2\|\mathbf{T}_2\| >$

$\eta/2\} \to 0$. □

Theorem 4.28 (Null distribution of the Breusch–Pagan statistic; FREES §5.7.1)**.** Let $\varepsilon_1, \varepsilon_2, \dots$ be independent and identically distributed with mean 0, variance $\sigma^2 > 0$, $\mathsf{E}[\varepsilon_1^4] < \infty$ and $\kappa = \operatorname{var}(\varepsilon_1^2)/\sigma^4 > 0$. For every n let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ has rank $k+1$ and rows $\mathbf{x}_i$, and $\mathbf{Z}$ has rows $\mathbf{z}_i \in \mathbb{R}^p$, both the first n terms of fixed sequences with $\|\mathbf{x}_i\|, \|\mathbf{z}_i\| \leqslant B$; let $\mathbf{Z}_1 = [\mathbf{1}\ \mathbf{Z}]$ have rank $p+1$, and let $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X$ and $n^{-1}\mathbf{Z}_1'\mathbf{Z}_1 \to \mathbf{Q}_1$, both positive definite. Then the statistic LM of Definition 4.26, with s^2 or with $\tilde{\sigma}^2$, satisfies $(2/\kappa)\,\mathrm{LM} \xrightarrow{d} \chi^2_p$. Under A5, $\kappa = 2$ and $\mathrm{LM} \xrightarrow{d} \chi^2_p$.

Proof. Write $g_i = \varepsilon_i^2/\sigma^2 - 1$, $\bar{\mathbf{z}}_n = n^{-1}\mathbf{Z}'\mathbf{1}$ and $\tilde{\mathbf{Z}} = \mathbf{Z} - \mathbf{1}\bar{\mathbf{z}}_n'$, whose columns sum to zero.

Step 1: a central limit theorem. The g_i are independent and identically distributed with mean 0 and variance $\kappa > 0$, and $\sup_i \mathsf{E}[g_i^2; |g_i| > a] = \mathsf{E}[g_1^2; |g_1| > a] \to 0$ as $a \to \infty$ by dominated convergence. The vectors $\mathbf{w}_i = (1, \mathbf{z}_i')'$ satisfy $n^{-1}\sum_i \mathbf{w}_i\mathbf{w}_i' = n^{-1}\mathbf{Z}_1'\mathbf{Z}_1 \to \mathbf{Q}_1$ and $n^{-1}\max_i \|\mathbf{w}_i\|^2 \leqslant (1 + B^2)/n \to 0$. The prerequisites' corollary **weighted sums** gives $n^{-1/2}\mathbf{Z}_1'\mathbf{g} \xrightarrow{d} N(\mathbf{0}, \kappa\mathbf{Q}_1)$. Now $\tilde{\mathbf{Z}} = \mathbf{Z}_1\mathbf{L}_n'$ with $\mathbf{L}_n = [-\bar{\mathbf{z}}_n \ \ \mathbf{I}_p]$; $\bar{\mathbf{z}}_n$ is part of the first column of $n^{-1}\mathbf{Z}_1'\mathbf{Z}_1$, so it converges to some $\boldsymbol{\mu}$, and $\mathbf{L}_n \to \mathbf{L} = [-\boldsymbol{\mu} \ \ \mathbf{I}_p]$. By the prerequisites' Slutsky

theorem, part (ii), with $\Phi(\mathbf{v}, \mathbf{L}) = \mathbf{L}\mathbf{v}$, and their theorem **closure under affine maps**,

$$n^{-1/2}\tilde{\mathbf{Z}}'\mathbf{g} = \mathbf{L}_n\, n^{-1/2}\mathbf{Z}_1'\mathbf{g} \xrightarrow{d} N(\mathbf{0}, \kappa\mathbf{Q}), \qquad \mathbf{Q} = \mathbf{L}\mathbf{Q}_1\mathbf{L}' = \lim_n n^{-1}\tilde{\mathbf{Z}}'\tilde{\mathbf{Z}},$$

and $\mathbf{Q}$ is positive definite, since $\mathbf{L}'\mathbf{v} \neq \mathbf{0}$ for $\mathbf{v} \neq \mathbf{0}$.

Step 2: residuals for errors. With $\hat{g}_i = e_i^2/\sigma^2 - 1$, Lemma 4.27 with $\mathbf{a}_{ni} = (\mathbf{z}_i - \bar{\mathbf{z}}_n)/\sigma^2$, of norm at most $2B/\sigma^2$, gives $n^{-1/2}\tilde{\mathbf{Z}}'(\hat{\mathbf{g}} - \mathbf{g}) \xrightarrow{p} \mathbf{0}$, so $n^{-1/2}\tilde{\mathbf{Z}}'\hat{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \kappa\mathbf{Q})$ by Slutsky's theorem.

Step 3: the variance estimate. $\tilde{\sigma}^2 = n^{-1}\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} = n^{-1}\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} - n^{-1}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$. By Chebyshev's inequality with $\operatorname{var}(n^{-1}\sum_i \varepsilon_i^2) = \kappa\sigma^4/n$, $n^{-1}\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} \xrightarrow{p} \sigma^2$; and $n^{-1}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \geqslant 0$ has mean $\sigma^2(k+1)/n$, so it tends to 0 in probability by Markov's inequality. Hence $\tilde{\sigma}^2 \xrightarrow{p} \sigma^2$, and $s^2 = \tilde{\sigma}^2\, n/(n-k-1) \xrightarrow{p} \sigma^2$. Let $\hat{\sigma}^2$ be either and $u_i = e_i^2/\hat{\sigma}^2$. Since $\tilde{\mathbf{Z}}'\mathbf{1} = \mathbf{0}$, $\tilde{\mathbf{Z}}'(\mathbf{u} - \mathbf{1}) = \tilde{\mathbf{Z}}'\mathbf{u} = (\sigma^2/\hat{\sigma}^2)\,\tilde{\mathbf{Z}}'\hat{\mathbf{g}}$, and $\sigma^2/\hat{\sigma}^2 \xrightarrow{p} 1$ by the prerequisites' continuous mapping theorem. By Slutsky's theorem, $\mathbf{V}_n = n^{-1/2}\tilde{\mathbf{Z}}'\mathbf{u} \xrightarrow{d} N(\mathbf{0}, \kappa\mathbf{Q})$.

Step 4: the statistic. By (2) for the design $\mathbf{Z}_1$, the regression sum of squares of step (iii) is

$\|(\mathbf{H}_{\mathbf{Z}_1} - \frac{1}{n}\mathbf{1}\mathbf{1}')\mathbf{u}\|^2 = \mathbf{u}'\tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}'\tilde{\mathbf{Z}})^{-1}\tilde{\mathbf{Z}}'\mathbf{u}$, so

$$\frac{2}{\kappa}\mathrm{LM} = \mathbf{V}_n'\left(\kappa\, n^{-1}\tilde{\mathbf{Z}}'\tilde{\mathbf{Z}}\right)^{-1}\mathbf{V}_n.$$

$\kappa n^{-1}\tilde{\mathbf{Z}}'\tilde{\mathbf{Z}}$ is symmetric and tends to $\kappa\mathbf{Q}$, positive definite, and the prerequisites' theorem **quadratic forms of asymptotically normal vectors** gives $(2/\kappa)\mathrm{LM} \xrightarrow{d} \chi^2_p$. Under A5, $\varepsilon_1^2/\sigma^2 \sim \chi^2_1$, whose variance is 2 by the prerequisites' moments of χ^2. □

Proposition 4.29 (Versions of the statistic; FREES §5.7.1)**.** Under A4, let $\mathbf{Z}_1 = [\mathbf{1}\ \mathbf{Z}]$ have rank $p+1$ and SSE > 0, let LM_s and $\mathrm{LM}_{\tilde{\sigma}}$ be the statistic of Definition 4.26 with s^2 and with $\tilde{\sigma}^2$, let $\hat{\tau}^2 = n^{-1}\sum_i (e_i^2 - \tilde{\sigma}^2)^2 > 0$, and let R_z^2 be the coefficient of determination of the regression of e_i^2 on $\mathbf{z}_i$ with an intercept. Then

$$\mathrm{LM}_s = \left(\frac{n-k-1}{n}\right)^2 \mathrm{LM}_{\tilde{\sigma}}, \qquad \mathrm{LM}_{\tilde{\sigma}} = nR_z^2 \cdot \frac{\hat{\tau}^2}{2\tilde{\sigma}^4}.$$

Proof. The regression sum of squares is a quadratic form in the response, so multiplying the response by c multiplies it by c^2; and $e_i^2/s^2 = \{(n-k-1)/n\}\, e_i^2/\tilde{\sigma}^2$. For the second identity let $v_i = e_i^2$, with regression and total sums of squares SSR_v and SST_v. Then $\mathrm{LM}_{\tilde{\sigma}} = \mathrm{SSR}_v/(2\tilde{\sigma}^4)$; $\bar{v} = \mathrm{SSE}/n = \tilde{\sigma}^2$, so $\mathrm{SST}_v = n\hat{\tau}^2$; and $R_z^2 = \mathrm{SSR}_v/\mathrm{SST}_v$. Hence $nR_z^2\hat{\tau}^2/(2\tilde{\sigma}^4) = \mathrm{SSR}_v/(2\tilde{\sigma}^4)$. □

Corollary 4.30 (Koenker's studentised statistic (beyond the syllabus))**.** Under the hypotheses of Theorem 4.28 with $\mathsf{E}[\varepsilon_1^8] < \infty$, $nR_z^2 \xrightarrow{d} \chi_p^2$, whatever the value of κ.

Proof. By Proposition 4.29, $nR_z^2 = (2/\kappa)\mathrm{LM}_{\tilde{\sigma}} \cdot \kappa\tilde{\sigma}^4/\hat{\tau}^2$, and the first factor tends in distribution to χ_p^2 by Theorem 4.28. By Slutsky's theorem it suffices that $\hat{\tau}^2/\tilde{\sigma}^4 \xrightarrow{p} \kappa$. Since $n^{-1}\sum_i e_i^2 = \tilde{\sigma}^2$, $\hat{\tau}^2 = n^{-1}\sum_i e_i^4 - \tilde{\sigma}^4$, and $\tilde{\sigma}^4 \xrightarrow{p} \sigma^4$ (Step 3 of Theorem 4.28 and continuous mapping); it remains that $n^{-1}\sum_i e_i^4 \xrightarrow{p} \mu_4 = \mathsf{E}[\varepsilon_1^4] = (\kappa+1)\sigma^4$. First, $n^{-1}\sum_i \varepsilon_i^4 \xrightarrow{p} \mu_4$ by Chebyshev's inequality, since $\mathrm{var}(\varepsilon_1^4) \leqslant \mathsf{E}[\varepsilon_1^8] < \infty$. Next, with $d_i = \mathbf{x}_i'\boldsymbol{\delta}$ as in Lemma 4.27, $|d_i| \leqslant B\|\boldsymbol{\delta}\| =: \xi_n$ and

$e_i^4 - \varepsilon_i^4 = -4\varepsilon_i^3 d_i + 6\varepsilon_i^2 d_i^2 - 4\varepsilon_i d_i^3 + d_i^4$, so

$$\left|\frac{1}{n}\sum_i (e_i^4 - \varepsilon_i^4)\right| \leqslant 4\xi_n m_3 + 6\xi_n^2 m_2 + 4\xi_n^3 m_1 + \xi_n^4, \qquad m_r = \frac{1}{n}\sum_i |\varepsilon_i|^r.$$

$\mathsf{E}[m_r] = \mathsf{E}\,|\varepsilon_1|^r \leqslant 1 + \mu_4$, and $\mathsf{E}[\xi_n^2] \leqslant B^2\sigma^2 \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}) = B^2\sigma^2 n^{-1} \operatorname{tr}((n^{-1}\mathbf{X}'\mathbf{X})^{-1}) \to 0$, the entries of $(n^{-1}\mathbf{X}'\mathbf{X})^{-1}$ being bounded, so $\xi_n \xrightarrow{p} 0$. A product $U_n V_n$ with $U_n \xrightarrow{p} 0$ and $\mathsf{E}\,|V_n| \leqslant K$ tends to 0 in probability: $\mathsf{P}\{|U_n V_n| > \eta\} \leqslant K/M + \mathsf{P}\{|U_n| > \eta/M\}$ for every $M > 0$. Each term of the bound therefore tends to 0 in probability, and $\hat\tau^2/\tilde\sigma^4 \xrightarrow{p} (\mu_4 - \sigma^4)/\sigma^4 = \kappa$. □

Example 4.15 (Testing the funnel; FREES §5.7.1)**.** With $\mathbf{z}_i$ the two explanatory variables, FREES's five steps with s^2 and with $\tilde\sigma^2$, against `bptest`, for `mpg` and for ln `mpg`:

```
library(lmtest)
bp_versions <- function(m) {
  e <- residuals(m); nn <- length(e); kk <- length(coef(m)) - 1
  Z <- model.matrix(m)[, -1, drop = FALSE]; s2t <- sum(e^2) / nn; s2 <- sum(e^2) /
   ↪  (nn - kk - 1)
  ssr <- function(u) sum((fitted(lm(u ~ Z)) - mean(u))^2)
```

```
  c(LM_s = ssr(e^2 / s2) / 2, LM_tilde = ssr(e^2 / s2t) / 2,
    bptest_studentize_FALSE = unname(bptest(m, studentize = FALSE)$statistic),
    nR2 = nn * summary(lm(e^2 ~ Z))$r.squared, bptest_default =
     ↪ unname(bptest(m)$statistic),
    tau2_over_2sigma4 = mean((e^2 - s2t)^2) / (2 * s2t^2), factor_s = ((nn - kk - 1)
     ↪ / nn)^2,
    p_LM_tilde = pchisq(ssr(e^2 / s2t) / 2, ncol(Z), lower.tail = FALSE),
    p_nR2 = pchisq(nn * summary(lm(e^2 ~ Z))$r.squared, ncol(Z), lower.tail =
     ↪ FALSE))
}
round(rbind(mpg = bp_versions(a2), log_mpg = bp_versions(l2)), 4)
```

```
##           LM_s LM_tilde bptest_studentize_FALSE    nR2
## mpg     56.086   56.955                 56.955 34.528
## log_mpg  7.355    7.469                  7.469  5.202
##         bptest_default tau2_over_2sigma4 factor_s p_LM_tilde
## mpg             34.528            1.649   0.9848     0.0000
## log_mpg          5.202            1.436   0.9848     0.0239
##          p_nR2
## mpg     0.0000
## log_mpg 0.0742
```

For `mpg`, $\mathrm{LM}_{\tilde\sigma} = 56.95$, which `bptest(…, studentize = FALSE)` reproduces, and $\mathrm{LM}_s = 56.09$ is smaller by the factor 0.985 of Proposition 4.29; either rejects homoscedasticity at any usual level against χ^2_2. `bptest`'s default is $nR^2_z = 34.53$, the ratio $\hat\tau^2/(2\tilde\sigma^4) = 1.65$ being the measured departure of the fourth moment of the residuals from the normal value. For $\ln$ `mpg` the statistic falls to 7.47 ($p = 0.024$) and nR^2_z to 5.20 ($p = 0.074$).

Example 4.16 (The null distribution; FREES §5.7.1)**.** Theorem 4.28 says that LM tends to $(\kappa/2)\chi^2_p$, so its test has asymptotic level $\mathsf{P}\{\chi^2_p > (2/\kappa)\chi^2_{p,0.95}\}$, which is 0.05 only when $\kappa = 2$. Simulated rejection rates at the nominal 5% with $n = 200$, $p = 1$, and errors that are normal ($\kappa = 2$), centred exponential ($\kappa = 8$) and uniform ($\kappa = 0.8$):

```
set.seed(2); nb <- 200; xb <- runif(nb); Xb <- cbind(1, xb)
Mb <- diag(nb) - Xb %*% solve(crossprod(Xb)) %*% t(Xb); zc <- xb - mean(xb)
gen <- list(normal = function() rnorm(nb), exponential = function() rexp(nb) - 1,
            uniform = function() runif(nb, -sqrt(3), sqrt(3)))
kap <- c(normal = 2, exponential = 8, uniform = 0.8)
round(t(sapply(names(gen), function(g) {
  st <- replicate(2000, {
    e <- drop(Mb %*% gen[[g]]()); u <- e^2 / mean(e^2)
```

```
    c(LM = sum(zc * u)^2 / sum(zc^2) / 2,
      nR2 = nb * sum(zc * e^2)^2 / sum(zc^2) / sum((e^2 - mean(e^2))^2))
  })
  c(kappa = kap[[g]], LM_reject = mean(st[1, ] > qchisq(0.95, 1)),
    limit = pchisq(qchisq(0.95, 1) * 2 / kap[[g]], 1, lower.tail = FALSE),
    nR2_reject = mean(st[2, ] > qchisq(0.95, 1)))
})), 3)
```

```
##             kappa LM_reject limit nR2_reject
## normal        2.0     0.046 0.050      0.053
## exponential   8.0     0.294 0.327      0.050
## uniform       0.8     0.004 0.002      0.058
```

The statistic LM rejects a true null 4.6% of the time with normal errors, 29% with exponential errors (limit 33%) and 0.4% with uniform errors (limit 0.2%); Koenker's nR_z^2 stays near 5% in all three.

5.2 Heteroscedasticity-consistent standard errors

Definition 4.31 (Robust standard errors; FREES §5.7.2)**.** With weights $c_i > 0$, let

$$\hat{\mathbf{V}}_c = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{n} c_i\, e_i^2 \mathbf{x}_i\mathbf{x}_i'\Big)(\mathbf{X}'\mathbf{X})^{-1}, \qquad \mathrm{SE}_r(\hat\beta_j) = \sqrt{[\hat{\mathbf{V}}_c]_{jj}}.$$

FREES's **empirical** or **robust** estimator is HC0, $c_i = 1$; HC1 takes $c_i = n/(n-k-1)$, HC2 $c_i = 1/(1-h_{ii})$ and HC3 $c_i = 1/(1-h_{ii})^2$, the last two built from the standardised and the deleted residuals ($h_{ii} < 1$).

Proposition 4.32 (Expected squared residuals; FREES §5.7.2)**.** Under A1, A2 and A4 with uncorrelated errors, $\mathrm{var}(\varepsilon_i) = \sigma_i^2$,

$$\mathsf{E}[e_i^2] = (1-2h_{ii})\sigma_i^2 + \sum_{j=1}^{n} h_{ij}^2\sigma_j^2.$$

Under homoscedasticity $\mathsf{E}[e_i^2] = (1-h_{ii})\sigma^2$, so $\mathsf{E}[e_i^2/(1-h_{ii})] = \sigma^2$: the middle matrix of HC2 is

then unbiased for $\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$, that of HC0 is biased downward, and that of HC3 upward.

Proof. $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$, so $e_i = \sum_j m_{ij}\varepsilon_j$ with $m_{ii} = 1 - h_{ii}$ and $m_{ij} = -h_{ij}$, and with mean-zero uncorrelated errors $\mathsf{E}[e_i^2] = \sum_j m_{ij}^2 \sigma_j^2 = (1-h_{ii})^2\sigma_i^2 + \sum_{j\neq i} h_{ij}^2 \sigma_j^2$. Expanding $(1-h_{ii})^2 = 1 - 2h_{ii} + h_{ii}^2$ and moving $h_{ii}^2\sigma_i^2$ into the sum gives the formula. With $\sigma_j^2 = \sigma^2$, $\sum_j h_{ij}^2 = h_{ii}$ (from $\mathbf{H} = \mathbf{H}^2$, Unit 2's theorem **properties of leverage**) gives $\sigma^2(1-h_{ii})$. Then $\mathsf{E}[e_i^2] = (1-h_{ii})\sigma^2 \leqslant \sigma^2 \leqslant \sigma^2/(1-h_{ii}) = \mathsf{E}[e_i^2/(1-h_{ii})^2]$, and each middle matrix is a sum of the corresponding multiples of $\mathbf{x}_i\mathbf{x}_i'$. □

Theorem 4.33 (Consistency of the robust standard errors; FREES §5.7.2)**.** For every n let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of rank $k+1$ and rows $\mathbf{x}_i$ from a fixed sequence, $\|\mathbf{x}_i\| \leqslant B$; let the errors be independent with mean 0, $\operatorname{var}(\varepsilon_i) = \sigma_i^2$ and $\sup_i \mathsf{E}[\varepsilon_i^4] \leqslant C$; and let $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X$ and $n^{-1}\sum_i \sigma_i^2 \mathbf{x}_i\mathbf{x}_i' \to \mathbf{F}$, both positive definite. Then

(i) $n^{-1}\sum_i e_i^2 \mathbf{x}_i\mathbf{x}_i' - n^{-1}\sum_i \sigma_i^2\mathbf{x}_i\mathbf{x}_i' \xrightarrow{p} \mathbf{0}$;

(ii) for every $\mathbf{c} \neq \mathbf{0}$, $(\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta})/\sqrt{\mathbf{c}'\hat{\mathbf{V}}_c\mathbf{c}} \xrightarrow{d} N(0,1)$ for HC0;

(iii) (ii) holds for HC1, HC2 and HC3.

Proof. **(i)** Split the difference into $n^{-1}\sum_i(e_i^2-\varepsilon_i^2)\mathbf{x}_i\mathbf{x}_i'$ and $n^{-1}\sum_i(\varepsilon_i^2-\sigma_i^2)\mathbf{x}_i\mathbf{x}_i'$. Since $\sigma_i^2 \leqslant 1+\mathsf{E}[\varepsilon_i^4] \leqslant 1+C$, Lemma 4.27 with $\mathbf{a}_{ni}$ the entries of $\mathbf{x}_i\mathbf{x}_i'$, of norm at most B^2, shows that $n^{1/2}$ times the first tends to $\mathbf{0}$ in probability, hence so does the first. Each entry of the second has mean 0 and variance at most $n^{-2}\sum_i \mathsf{E}[\varepsilon_i^4]B^4 \leqslant CB^4/n$, so it tends to 0 in probability by Chebyshev's inequality.

(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}) = (n^{-1}\mathbf{X}'\mathbf{X})^{-1}n^{-1/2}\sum_i \mathbf{x}_i\varepsilon_i$. The vectors $\mathbf{x}_i\varepsilon_i$ are independent with mean $\mathbf{0}$, $\mathsf{E}\,\|\mathbf{x}_i\varepsilon_i\|^4 \leqslant B^4C$ and $n^{-1}\sum_i \operatorname{var}(\mathbf{x}_i\varepsilon_i) = n^{-1}\sum_i \sigma_i^2\mathbf{x}_i\mathbf{x}_i' \to \mathbf{F}$, so the prerequisites' Lyapunov corollary with $\delta=2$ gives $n^{-1/2}\sum_i \mathbf{x}_i\varepsilon_i \xrightarrow{d} N(\mathbf{0},\mathbf{F})$. By the prerequisites' lemma **inverses of convergent positive definite matrices**, $(n^{-1}\mathbf{X}'\mathbf{X})^{-1} \to \mathbf{Q}_X^{-1}$, and Slutsky's theorem with the affine-map theorem gives $\sqrt{n}\,\mathbf{c}'(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}) \xrightarrow{d} N(0,v)$, $v = \mathbf{c}'\mathbf{Q}_X^{-1}\mathbf{F}\mathbf{Q}_X^{-1}\mathbf{c} > 0$. By (i) and continuous mapping, $n\,\mathbf{c}'\hat{\mathbf{V}}_c\mathbf{c} = \mathbf{c}'(n^{-1}\mathbf{X}'\mathbf{X})^{-1}(n^{-1}\sum_i e_i^2\mathbf{x}_i\mathbf{x}_i')(n^{-1}\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} \xrightarrow{p} v$, and the ratio tends to $N(0,1)$ by Slutsky's theorem.

(iii) For any symmetric $\mathbf{A}$ and vector $\mathbf{x}$, $|\mathbf{x}'\mathbf{A}\mathbf{x}| \leqslant \|\mathbf{x}\|^2\sum_{r,s}|a_{rs}|$; hence $\max_i h_{ii} \leqslant \eta_n = (B^2/n)\sum_{r,s}|[(n^{-1}\mathbf{X}'\mathbf{X})^{-1}]_{rs}| \to 0$, the entries being bounded by the same lemma. The weights of HC1 to HC3 lie between 1 and $\max\{n/(n-k-1),(1-\eta_n)^{-2}\} \to 1$, so the difference of their

middle matrices from that of HC0, divided by n, has entries at most $(\max_i c_i - 1)\, B^2\, \mathrm{SSE}/n$, where $\mathsf{E}[\mathrm{SSE}/n] \leqslant n^{-1}\sum_i \sigma_i^2 \leqslant 1 + C$ (Proposition 4.9); it tends to $\mathbf{0}$ in probability, and the argument of (ii) applies. □

Example 4.17 (Robust standard errors; FREES §5.7.2)**.** The four estimators from Definition 4.31 for the quadratic `mpg` fit, against `sandwich::vcovHC`, and the coverage of nominal 95% intervals for a slope when $\mathrm{sd}(\varepsilon_i) = x_i^2$ with x_i uniform on $(0, 3)$:

```
hc_manual <- function(m, type) {
  X <- model.matrix(m); e <- residuals(m); h <- hatvalues(m); XtXi <-
   ↪  solve(crossprod(X))
  cw <- switch(type, HC0 = 1, HC1 = nrow(X) / (nrow(X) - ncol(X)), HC2 = 1 / (1 -
   ↪  h), HC3 = 1 / (1 - h)^2)
  sqrt(diag(XtXi %*% crossprod(X * sqrt(cw) * e) %*% XtXi))
}
types <- c("HC0", "HC1", "HC2", "HC3")
se_tab <- cbind(usual = sqrt(diag(vcov(a2))), sapply(types, function(tp)
 ↪  hc_manual(a2, tp)))
round(se_tab, 6)
```

```
##                   usual      HC0      HC1     HC2      HC3
## (Intercept)     1.800427 2.074212 2.082195 2.09025 2.106535
## horsepower      0.031125 0.032966 0.033093 0.03323 0.033501
## I(horsepower^2) 0.000122 0.000119 0.000119 0.00012 0.000121
```

```
round(se_tab[, types] / se_tab[, "usual"], 3)
```

```
##                   HC0   HC1   HC2   HC3
## (Intercept)     1.152 1.157 1.161 1.170
## horsepower      1.059 1.063 1.068 1.076
## I(horsepower^2) 0.971 0.975 0.979 0.988
```

```
c(max_diff_sandwich = max(sapply(types, function(tp)
  max(abs(hc_manual(a2, tp) - sqrt(diag(sandwich::vcovHC(a2, type = tp))))))))
```

```
## max_diff_sandwich
##         4.663e-14
```

```
set.seed(3)
cover <- function(nn, R = 4000) {
  x <- runif(nn, 0, 3); X <- cbind(1, x); XtXi <- solve(crossprod(X)); h <-
   ↪  rowSums((X %*% XtXi) * X)
  res <- replicate(R, {
```

```
    y <- 1 + 2 * x + x^2 * rnorm(nn); b <- XtXi %*% crossprod(X, y); e <- drop(y - X
    ↪  %*% b)
    V0 <- XtXi %*% crossprod(X * e) %*% XtXi; V3 <- XtXi %*% crossprod(X * (e / (1 -
    ↪  h))) %*% XtXi
    q <- qt(0.975, nn - 2)
    c(usual = abs(b[2] - 2) <= q * sqrt(sum(e^2) / (nn - 2) * XtXi[2, 2]),
      HC0 = abs(b[2] - 2) <= q * sqrt(V0[2, 2]), HC3 = abs(b[2] - 2) <= q *
      ↪  sqrt(V3[2, 2]))
  })
  c(n = nn, max_h = max(h), rowMeans(res))
}
round(t(sapply(c(20, 50, 200, 1000), cover)), 3)
```

```
##         n max_h usual   HC0   HC3
## [1,]   20 0.196 0.897 0.889 0.939
## [2,]   50 0.088 0.878 0.922 0.939
## [3,]  200 0.019 0.889 0.941 0.944
## [4,] 1000 0.004 0.896 0.949 0.950
```

For the quadratic fit the robust standard errors exceed the usual ones by 15% to 17% for the intercept and 6% to 8% for the linear term. In the simulation the usual interval covers the slope

88% to 90% of the time at every n, since it estimates the wrong variance; the HC0 interval rises from 89% to the nominal level as $\max_i h_{ii}$ falls, as Theorem 4.33 predicts, and HC3 is close to 95% already at $n = 20$, in line with FREES's advice to prefer an estimator built from rescaled residuals.

5.3 Weighted least squares

Theorem 4.34 (Weighted least squares; FREES §5.7.3, ISLR §3.3.3)**.** Assume A1, A2, A4 and $\operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1}$ with $\sigma^2 > 0$, $\mathbf{W} = \operatorname{diag}(w_1, \dots, w_n)$ and known $w_i > 0$. Let $\mathbf{Y}^* = \mathbf{W}^{1/2}\mathbf{Y}$ and $\mathbf{X}^* = \mathbf{W}^{1/2}\mathbf{X}$, the rows multiplied by $\sqrt{w_i}$.

(i) Least squares on $(\mathbf{X}^*, \mathbf{Y}^*)$ gives $\hat{\boldsymbol{\beta}}_W = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y}$, the minimiser of $\sum_i w_i (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2$.

(ii) $\hat{\boldsymbol{\beta}}_W$ is unbiased with $\operatorname{var}(\hat{\boldsymbol{\beta}}_W) = \sigma^2 (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$ and is the best linear unbiased estimator of $\boldsymbol{\beta}$.

(iii) $s_W^2 = \sum_i w_i (y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}_W)^2/(n-k-1)$ is unbiased for σ^2, and with normal errors the t procedures of Unit 2 and the general linear hypothesis F test of Unit 3 applied to $(\mathbf{X}^*, \mathbf{Y}^*)$ are exact; the test of $\beta_1 = \cdots = \beta_k = 0$ is the latter with $\mathbf{R} = [\mathbf{0}\ \mathbf{I}_k]$, since $\mathbf{X}^*$ has no constant column.

Proof. $\mathbf{Y}^* = \mathbf{X}^*\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*$ with $\boldsymbol{\varepsilon}^* = \mathbf{W}^{1/2}\boldsymbol{\varepsilon}$, $\mathsf{E}[\boldsymbol{\varepsilon}^*] = \mathbf{0}$ and, by the prerequisites' proposition **variance under a linear map**, $\mathrm{var}(\boldsymbol{\varepsilon}^*) = \mathbf{W}^{1/2}\sigma^2\mathbf{W}^{-1}\mathbf{W}^{1/2} = \sigma^2\mathbf{I}_n$; $\mathbf{X}^*$ has full column rank since $\mathbf{W}^{1/2}$ is invertible. So $(\mathbf{X}^*, \mathbf{Y}^*)$ satisfies A1–A4 except that its first column is $(\sqrt{w_1}, \ldots, \sqrt{w_n})'$ rather than $\mathbf{1}$; the theorems of Units 2 and 3 cited in this proof use only $\mathrm{rank}(\mathbf{X}^*) = k+1$ and the moments, or the normality, of $\boldsymbol{\varepsilon}^*$, not a constant column. (i) Its least squares estimator is $(\mathbf{X}^{*\prime}\mathbf{X}^*)^{-1}\mathbf{X}^{*\prime}\mathbf{Y}^* = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y}$, and $\|\mathbf{Y}^* - \mathbf{X}^*\boldsymbol{\beta}\|^2 = \sum_i w_i(y_i - \mathbf{x}_i'\boldsymbol{\beta})^2$. (ii) Unit 2's theorems **unbiasedness**, **variance–covariance matrix** and **Gauss–Markov** for the starred model; an estimator $\mathbf{C}\mathbf{Y}$ linear in $\mathbf{Y}$ is $\mathbf{C}\mathbf{W}^{-1/2}\mathbf{Y}^*$, linear in $\mathbf{Y}^*$, so the comparison class is the same. (iii) The residuals of the starred fit are $\mathbf{W}^{1/2}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_W)$, whose squared length is $\sum_i w_i(y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}_W)^2$; Unit 2's theorem **unbiasedness of the mean square error** applies. With $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{W}^{-1})$, $\boldsymbol{\varepsilon}^* \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$ by the prerequisites' theorem **closure under affine maps**, and A5 holds for the starred model, so Unit 2's theorem **the t statistic** and Unit 3's theorem **the general linear hypothesis** apply to $(\mathbf{X}^*, \mathbf{Y}^*)$. □

Corollary 4.35 (Averages of raw observations; ISLR §3.3.3)**.** Let observation $g = 1, \ldots, G$ be the average $\bar{y}_g$ of n_g raw responses $y_{g1}, \ldots, y_{gn_g}$ that share the row $\mathbf{x}_g'$ and satisfy $y_{gl} = \mathbf{x}_g'\boldsymbol{\beta} + \varepsilon_{gl}$

with uncorrelated errors of mean 0 and variance $\sigma^2 > 0$, and let the $G \times (k+1)$ matrix of the $\mathbf{x}_g'$ have rank $k+1$. Then $\operatorname{var}(\bar{y}_g) = \sigma^2/n_g$, so $w_g = n_g$; the weighted least squares coefficients on the G averages equal the least squares coefficients on the $N = \sum_g n_g$ raw observations; and the raw residual sum of squares is $\sum_g \sum_l (y_{gl} - \bar{y}_g)^2 + \sum_g n_g (\bar{y}_g - \mathbf{x}_g' \hat{\boldsymbol{\beta}})^2$.

Proof. $\operatorname{var}(\bar{y}_g) = n_g^{-2} \sum_l \sigma^2 = \sigma^2/n_g$, and averages of disjoint sets of uncorrelated errors are uncorrelated, so Theorem 4.34 applies with $w_g = n_g$. The raw design repeats $\mathbf{x}_g'$ n_g times, so $\mathbf{X}_{\text{raw}}'\mathbf{X}_{\text{raw}} = \sum_g n_g \mathbf{x}_g \mathbf{x}_g' = \mathbf{X}'\mathbf{W}\mathbf{X}$ and $\mathbf{X}_{\text{raw}}'\mathbf{Y}_{\text{raw}} = \sum_g \mathbf{x}_g \sum_l y_{gl} = \sum_g n_g \mathbf{x}_g \bar{y}_g = \mathbf{X}'\mathbf{W}\bar{\mathbf{Y}}$: the normal equations coincide. Finally $y_{gl} - \mathbf{x}_g' \hat{\boldsymbol{\beta}} = (y_{gl} - \bar{y}_g) + (\bar{y}_g - \mathbf{x}_g' \hat{\boldsymbol{\beta}})$, and the cross term vanishes on summing over l because $\sum_l (y_{gl} - \bar{y}_g) = 0$. □

Example 4.18 (Fuel efficiency by model year; ISLR §3.3.3, FREES §5.7.3)**.** The 392 cars of `Auto` fall into 13 model years. Weighted least squares on the yearly means with weights the counts, three ways, against least squares on the raw data:

```
ym <- aggregate(mpg ~ year, data = Auto, FUN = mean); ym$n <-
 ↪  as.vector(table(Auto$year)); nrow(ym)
```

```
## [1] 13
```

```r
w_lm <- lm(mpg ~ year, data = ym, weights = n)
Xg <- cbind(1, ym$year); Wg <- diag(ym$n)
w_formula <- drop(solve(t(Xg) %*% Wg %*% Xg, t(Xg) %*% Wg %*% ym$mpg))
w_scaled <- coef(lm(I(sqrt(n) * mpg) ~ 0 + I(sqrt(n)) + I(sqrt(n) * year), data =
 ↪  ym))
raw <- lm(mpg ~ year, data = Auto)
rbind(lm_weights = coef(w_lm), formula = w_formula, sqrt_w_scaled =
 ↪  unname(w_scaled),
      raw_data = coef(raw), unweighted_means = coef(lm(mpg ~ year, data = ym)))
```

```
##                  (Intercept)  year
## lm_weights           -70.01 1.230
## formula              -70.01 1.230
## sqrt_w_scaled        -70.01 1.230
## raw_data             -70.01 1.230
## unweighted_means     -68.72 1.216
```

```r
range(ym$n)
```

```
## [1] 26 40
```

```r
SS_within <- sum((Auto$mpg - ave(Auto$mpg, Auto$year))^2)
c(SSE_raw = sum(residuals(raw)^2), within_plus_weighted = SS_within + sum(ym$n *
  residuals(w_lm)^2))
```

```
##              SSE_raw within_plus_weighted
##                15791                15791
```

The three weighted computations and the raw regression agree, while unweighted least squares on the means, which treats a year of 26 cars like a year of 40, gives a different line.

5.4 Transformations

Proposition 4.36 (Concave transformations and a spread that grows with the mean; FREES §5.7.4, ISLR §3.3.3)**.** Let h be increasing and convex on an interval I, let ε be a random variable, and for each μ in an interval J with $\mu + \varepsilon \in I$ almost surely let $Y_\mu = h(\mu + \varepsilon)$ with $\mathsf{E}[Y_\mu^2] < \infty$. Then $\mathsf{E}[Y_\mu]$ and $\operatorname{var}(Y_\mu)$ are nondecreasing in $\mu \in J$.

Proof. $\mathsf{E}[Y_\mu]$ is nondecreasing because h is increasing. Let ε' be an independent copy of ε; then $\operatorname{var}(Y_\mu) = \frac{1}{2}\,\mathsf{E}[\{h(\mu+\varepsilon) - h(\mu+\varepsilon')\}^2]$, by expanding the square. Fix $a < b$ and $\mu < \mu'$ in J with

the four points below in I, and put $x_1 = \mu + a$, $x_2 = \mu' + b$ and $\{y_1, y_2\} = \{\mu + b, \mu' + a\}$. Then $x_1 \leqslant y_j \leqslant x_2$ and $y_1 + y_2 = x_1 + x_2$, so $y_j = \lambda_j x_1 + (1 - \lambda_j)x_2$ with $\lambda_1 + \lambda_2 = 1$, and adding the two convexity inequalities $h(y_j) \leqslant \lambda_j h(x_1) + (1 - \lambda_j)h(x_2)$ gives $h(y_1) + h(y_2) \leqslant h(x_1) + h(x_2)$, that is

$$0 \leqslant h(\mu + b) - h(\mu + a) \leqslant h(\mu' + b) - h(\mu' + a),$$

the left inequality because h is increasing. Hence $|h(\mu + \varepsilon) - h(\mu + \varepsilon')|$ is nondecreasing in μ for every outcome, and taking expectations of its square completes the proof. □

Remark (FREES §5.7.4, ISLR §3.3.3)**.** If $g = h^{-1}$ is increasing and concave, as $\ln y$ is on $(0, \infty)$ and $\ln(1 + y)$ and $\sqrt{y}$ are on $[0, \infty)$, and $g(y_i) = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with identically distributed errors, then the variance of y_i does not decrease as its mean increases: that is the funnel that such a transformation of the response removes. An increasing convex g can only remove a spread that does not increase with the mean: $h = g^{-1}$ is then increasing and concave, and Proposition 4.36 applied to the increasing convex $t \mapsto -h(-t)$ and to $-\varepsilon$ makes $\mathrm{var}(y_i) = \mathrm{var}(-y_i)$ nonincreasing in the mean. A transformation also acts on the whole shape of the distribution, compressing the right tail when it is concave. $\ln(1 + y)$ admits zeros, and FREES's $100 \ln(1 + y/100)$ keeps

percentage data near their original units.

Remark (Limits of transformation; FREES §5.7.4)**.** When $g(y_i)$ has identically distributed errors, a concave or convex g (replace g by $-g$ if it is decreasing) cannot remove a spread that rises and then falls with the mean, since by Proposition 4.36 and the remark after it each can equalise only a spread that moves one way with the mean; such a pattern needs a transformation that is neither concave nor convex, such as the logit for proportions (concave below $\frac{1}{2}$, convex above). When the data are roughly symmetric but heteroscedastic, a transformation that equalises the spread is not linear, and so in general skews the distribution. Explanatory variables may be transformed too, since the regression assumptions condition on them: some analysts transform them towards normality, others avoid it because the fitted model is harder to interpret. FREES takes a middle course: it uses transformations that are readily interpreted, such as the logarithms, powers and indicators of its §3.5, and admits a transform that is hard to communicate only when it brings a substantial gain in fit or prediction.

Proposition 4.37 (Multiplicative errors; FREES §5.7.4)**.** Let $y_i = \exp(\mathbf{x}_i'\boldsymbol{\beta})\,\eta_i$, with an intercept in $\mathbf{x}_i$ and $\eta_1, \ldots, \eta_n$ positive, not almost surely constant, independent and identically distributed, $\mathsf{E}[\eta_1^2] < \infty$ and $\mathsf{E}[(\ln \eta_1)^2] < \infty$. Then $\mathrm{sd}(y_i)/\,\mathsf{E}[y_i] = \mathrm{sd}(\eta_1)/\,\mathsf{E}[\eta_1]$ for every i, while $\ln y_i = \mathbf{x}_i'\boldsymbol{\beta} + \ln \eta_i$ satisfies A1–A3 with intercept $\beta_0 + \mathsf{E}[\ln \eta_1]$ and errors $\ln \eta_i - \mathsf{E}[\ln \eta_1]$.

Proof. $\mathsf{E}[y_i] = e^{\mathbf{x}_i'\boldsymbol{\beta}}\,\mathsf{E}[\eta_1]$ and $\mathrm{sd}(y_i) = e^{\mathbf{x}_i'\boldsymbol{\beta}}\,\mathrm{sd}(\eta_1)$. The logarithm is $\mathbf{x}_i'\boldsymbol{\beta} + \mathsf{E}[\ln \eta_1] + (\ln \eta_i - \mathsf{E}[\ln \eta_1])$, with independent errors of mean 0 and common variance $\mathrm{var}(\ln \eta_1)$. □

Remark (FREES §5.7.4)**.** The standard deviation proportional to the mean is the funnel of Example 4.14, and the logarithm removes it exactly under Proposition 4.37; the Breusch–Pagan statistic of Example 4.15 fell from 56.95 to 7.47.

6 Computation in R

ISLR §3.6.2–3.6.3 ·FREES chapter 5 script.

6.1 The ISLR lab

`plot` applied to an `lm` fit draws four diagnostic plots:

```
lm.fit <- lm(medv ~ lstat, data = Boston)
op <- par(mfrow = c(2, 2), mar = c(3, 3.4, 1.8, 0.6), mgp = c(1.9, 0.6, 0))
plot(lm.fit, cex = 0.4)
par(op)
```

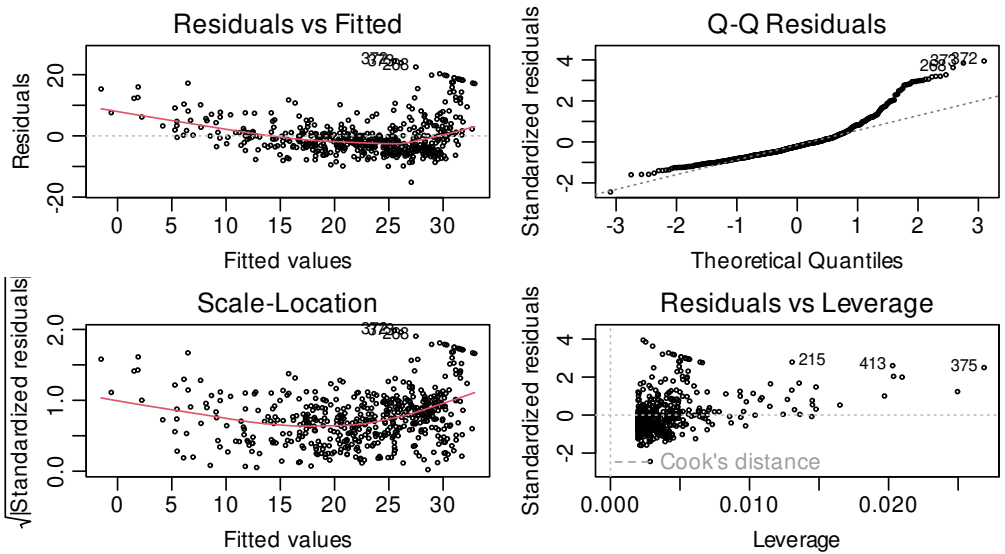
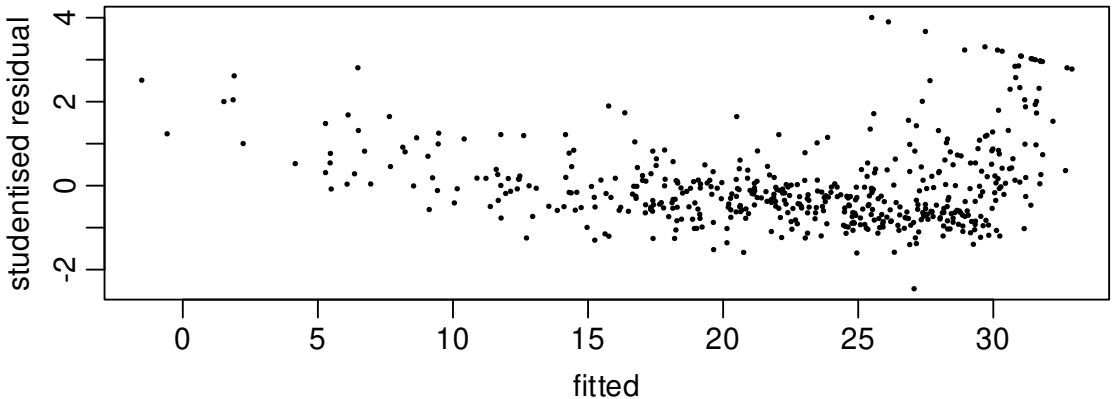


Figure 4.8: The four diagnostic plots of `medv` on `lstat`.

Residuals against fitted values show curvature, the check of Example 4.3; the normal quantile plot of the r_i checks A5; the scale–location plot, $\sqrt{|r_i|}$ against the fitted values, checks homoscedasticity; and the plot of r_i against h_{ii} draws, where they fall inside it, the contours on which Cook's distance equals 0.5 and 1, which by Theorem 4.14 are the curves $r = \pm\sqrt{(k+1)D(1-h)/h}$; here the largest Cook's distance is below 0.1 and no contour appears. The lab's residual plot of t_i, its largest leverage, and the Bonferroni test of Corollary 4.6 as `car::outlierTest` computes it:

```
op <- par(mar = c(3.2, 3.4, 0.6, 0.6), mgp = c(1.9, 0.6, 0))
plot(predict(lm.fit), rstudent(lm.fit), xlab = "fitted", ylab = "studentised residual",
 ↪  pch = 16, cex = 0.4)
par(op)
```




```r
c(which_max_h = which.max(hatvalues(lm.fit)), max_h = max(hatvalues(lm.fit)),
  max_D = max(cooks.distance(lm.fit)), n_abs_t_above_3 = sum(abs(rstudent(lm.fit)) >
   ↪  3))
```

```
## which_max_h.375          max_h          max_D
##       375.00000        0.02687        0.08616
## n_abs_t_above_3
##        11.00000
```

```r
t_max <- max(abs(rstudent(lm.fit))); n_B <- nobs(lm.fit)
c(t_max = t_max, bonferroni_p = min(1, n_B * 2 * pt(-t_max, n_B - 1 - 2)))
```

```
##        t_max bonferroni_p
##      4.00470      0.03616
```

```r
car::outlierTest(lm.fit)
```

```
##     rstudent unadjusted p-value Bonferroni p
## 372    4.005         0.00007147      0.03616
```

Eleven observations have $|t_i| > 3$, ISLR's flag, and the largest, $t = 4.00$ at observation 372, has Bonferroni p-value 0.036. The variance inflation factors of the full regression:

```r
lm.fit.all <- lm(medv ~ ., data = Boston)
round(car::vif(lm.fit.all), 2)
```

```
##    crim      zn   indus    chas     nox      rm     age
##    1.77    2.30    3.99    1.07    4.37    1.91    3.09
##     dis     rad     tax ptratio   lstat
##    3.95    7.45    9.00    1.80    2.87
```

All are below 10; the largest are **tax** (9.00) and **rad** (7.45).

6.2 FREES's chapter 5 script

```
op <- par(mfrow = c(2, 2), mar = c(3, 3.4, 1.8, 0.6), mgp = c(1.9, 0.6, 0))
plot(mv, cex = 0.4)
par(op)
```

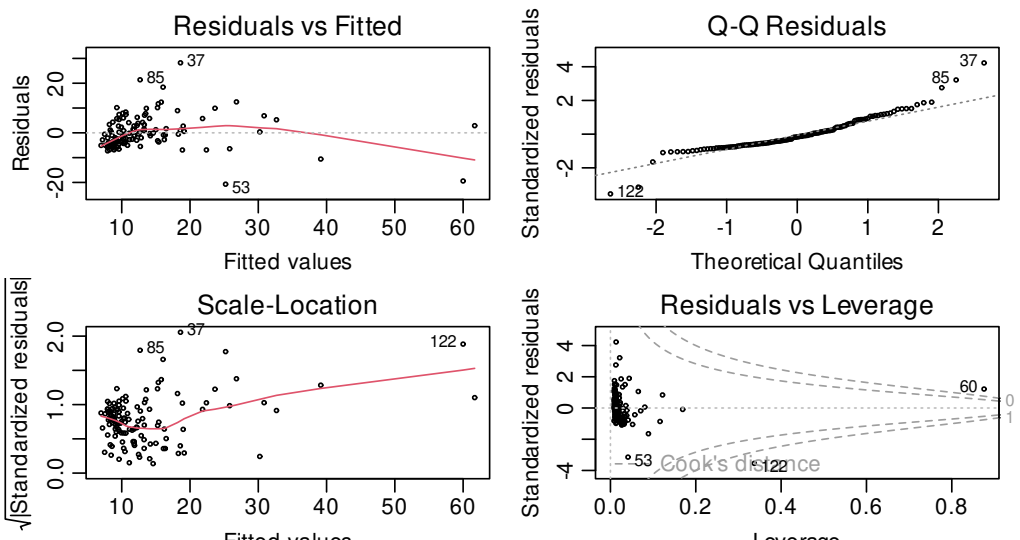


Figure 4.9: The four diagnostic plots of VOLUME on PRICE, SHARE and VALUE.

```r
c(hat_60 = hatvalues(mv)[[60]], residual_60 = residuals(mv)[[60]])
```

```
##      hat_60 residual_60
##      0.8756      2.8796
```

In the fourth panel IBM, observation 60, lies beyond the contour $D = 1$ at leverage 0.88 with a moderate residual; observation 122 is beyond $D = 1$ as well.

6.3 Formulas against package output

For the liquidity regression `mv`: the leverages from $\mathbf{H}$ and from the Mahalanobis form (3); r_i and t_i from Definition 4.4, from (1) and from the indicator regression of Proposition 4.3; the deleted coefficients from Proposition 4.3(iii) against refitting and `dfbeta`; Cook's distance from its definition and from Theorem 4.14; and Proposition 4.16.

```r
X <- model.matrix(mv); e <- residuals(mv); nX <- nrow(X); kX <- ncol(X) - 1
XtXi <- solve(crossprod(X)); H <- X %*% XtXi %*% t(X); h <- diag(H); s <-
 ↪  summary(mv)$sigma
Delta2 <- mahalanobis(X[, -1], colMeans(X[, -1]), cov(X[, -1]))
r <- e / (s * sqrt(1 - h))
s_del <- sqrt(((nX - kX - 1) * s^2 - e^2 / (1 - h)) / (nX - kX - 2))
```

```
t_i <- e / (s_del * sqrt(1 - h))
t_indicator <- sapply(c(37, 60, 122), function(i) {
  u <- as.numeric(seq_len(nX) == i); coef(summary(lm(VOLUME ~ PRICE + SHARE + VALUE +
  ↪  u, data = liq)))["u", 3]
})
signif(c(h_vs_hatvalues = max(abs(h - hatvalues(mv))), h_vs_mahalanobis = max(abs(h -
↪  (1 / nX + Delta2 / (nX - 1)))),
  r_vs_rstandard = max(abs(r - rstandard(mv))), t_vs_rstudent = max(abs(t_i -
  ↪  rstudent(mv))),
  t_from_r = max(abs(r * sqrt((nX - kX - 2) / (nX - kX - 1 - r^2)) - rstudent(mv))),
  t_indicator = max(abs(t_indicator - rstudent(mv)[c(37, 60, 122)]))), 3)
```

```
##   h_vs_hatvalues h_vs_mahalanobis   r_vs_rstandard
##         6.66e-16         8.88e-16         3.33e-15
##    t_vs_rstudent         t_from_r      t_indicator
##         3.55e-15         3.77e-15         1.78e-15
```

```
db_formula <- t(XtXi %*% t(X) %*% diag(e / (1 - h)))
db_refit <- t(sapply(seq_len(nX), function(i) coef(mv) - coef(lm(VOLUME ~ PRICE + SHARE
↪  + VALUE, data = liq[-i, ]))))
D_def <- sapply(seq_len(nX), function(i) sum((X %*% (coef(mv) - db_refit[i, ]) -
↪  fitted(mv))^2)) / ((kX + 1) * s^2)
```

```
signif(c(dfbeta_formula_vs_refit = max(abs(db_formula - db_refit)),
  dfbeta_vs_R = max(abs(db_formula - dfbeta(mv))),
  cook_definition_vs_R = max(abs(D_def - cooks.distance(mv))),
  cook_formula_vs_R = max(abs(r^2 / (kX + 1) * h / (1 - h) - cooks.distance(mv))),
  dffits_vs_R = max(abs(t_i * sqrt(h / (1 - h)) - dffits(mv))),
  dfbetas_vs_R = max(abs(db_formula / outer(s_del, sqrt(diag(XtXi))) - dfbetas(mv))),
  cook_from_dffits = max(abs(s_del^2 / ((kX + 1) * s^2) * dffits(mv)^2 -
   ↪  cooks.distance(mv)))), 3)
```

```
## dfbeta_formula_vs_refit            dfbeta_vs_R
##                1.35e-14               5.77e-15
##    cook_definition_vs_R      cook_formula_vs_R
##                1.55e-14               2.98e-14
##             dffits_vs_R           dfbetas_vs_R
##                1.91e-14               1.55e-14
##        cook_from_dffits
##                1.55e-15
```

Every difference is of the order of rounding error.

7 Exercises

Exercise 4.1 (FREES exercise 5.1)**.** In the regression on one explanatory variable, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$: (a) show that $h_{ii} = 1/n + (x_i - \bar{x})^2/\{(n-1)s_x^2\}$; (b) show that $\bar{h} = 2/n$; (c) if $h_{ii} = 6/n$, how many standard deviations is x_i from the mean?

Solution. (a) Proposition 4.12 with $k = 1$: $\mathbf{S} = s_x^2$ and $\Delta_i^2 = (x_i - \bar{x})^2/s_x^2$, so $h_{ii} = 1/n + (x_i - \bar{x})^2/\{(n-1)s_x^2\}$. (b) $\sum_i (x_i - \bar{x})^2 = (n-1)s_x^2$, so $\sum_i h_{ii} = 1 + 1 = 2$ and $\bar{h} = 2/n$, as Unit 2's theorem **properties of leverage** gives through $\text{tr}(\mathbf{H}) = k + 1$. (c) $6/n = 1/n + (x_i - \bar{x})^2/\{(n-1)s_x^2\}$ gives $|x_i - \bar{x}|/s_x = \sqrt{5(n-1)/n}$:

```
1 round(c(n20 = sqrt(5 * 19 / 20), n100 = sqrt(5 * 99 / 100), limit = sqrt(5)), 4)
```

```
##   n20  n100 limit
## 2.179 2.225 2.236
```

2.18 standard deviations at $n = 20$, increasing to $\sqrt{5} = 2.24$. The leverage $6/n$ is three times the mean, FREES's high leverage cutoff, so for $k = 1$ that cutoff marks points about 2.2 standard deviations from the mean. ■

Exercise 4.2 (FREES exercise 5.2)**.** A regression on one explanatory variable with $n = 3$ has residuals 3.181, −6.362, 3.181 and leverages 0.8333, 0.3333, 0.8333. Compute the **PRESS** statistic $\sum_i e_{(i)}^2$, the sum of squared deleted residuals.

Solution. By Proposition 4.3(iii), $e_{(i)} = e_i/(1 - h_{ii})$, so no refitting is needed:

```
e3 <- c(3.181, -6.362, 3.181)
c(PRESS_printed_h = sum((e3 / (1 - c(0.8333, 0.3333, 0.8333)))^2),
  PRESS_exact_h = sum((e3 / (1 - c(5, 2, 5) / 6))^2), SSE = sum(e3^2),
  ratio = sum((e3 / (1 - c(0.8333, 0.3333, 0.8333)))^2) / sum(e3^2))
```

```
## PRESS_printed_h   PRESS_exact_h             SSE
##          819.32          819.62           60.71
##           ratio
##           13.50
```

$PRESS = 819.3$ with the printed leverages; they are $5/6, 1/3, 5/6$, the leverages of $x = (-1, 0, 1)$ by Exercise 4.1(a), which give 819.6. $PRESS$ is $n\,\mathrm{CV}_{(n)}$ of Unit 1, and it is 13.5 times SSE = 60.7 here because the two outer points have $1 - h_{ii} = 1/6$. ■

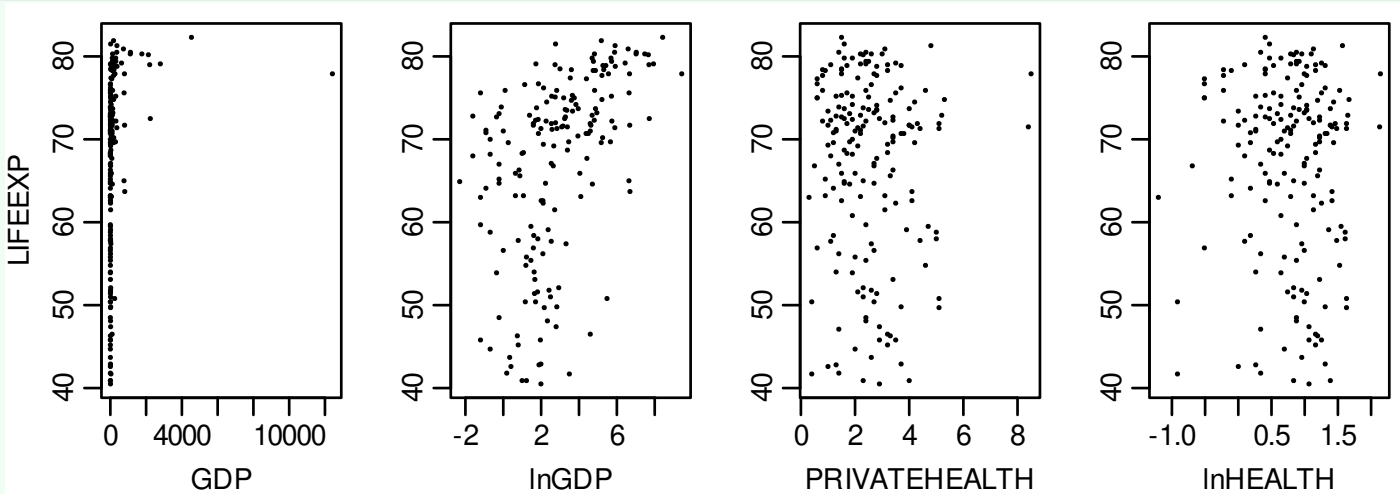
Exercise 4.3 (FREES exercise 5.3, parts (a), (c)). `UNLifeExpectancy`. (a) Plot life expectancy against the gross domestic product and against private expenditure on health; why are the logarithms `lnGDP` and `lnHEALTH` desirable? (c) For the regression of `LIFEEXP` on `FERTILITY`, `PUBLICEDUCATION` and `lnHEALTH`: (i) give histograms of the standardised residuals and leverages; (ii) find the standardised residual and leverage of Lesotho: an outlier, a high leverage point, or both? (iii) refit without Lesotho and compare.

Solution. (a)

```
un <- read.csv("data/frees/csv/UNLifeExpectancy.csv")
un$COUNTRY <- gsub(" ", " ", iconv(un$COUNTRY, "latin1", "UTF-8"))
un$lnGDP <- log(un$GDP); un$lnHEALTH <- log(un$PRIVATEHEALTH)
vs <- c("GDP", "lnGDP", "PRIVATEHEALTH", "lnHEALTH")
round(rbind(skewness = sapply(un[vs], function(x) skew(na.omit(x))),
            cor_LIFEEXP = sapply(un[vs], function(x) cor(x, un$LIFEEXP, use =
            ↪  "complete.obs"))), 3)
```

```
##                GDP lnGDP PRIVATEHEALTH lnHEALTH
## skewness     9.261 0.229         1.161   -0.685
## cor_LIFEEXP  0.213 0.476        -0.037   -0.033
```

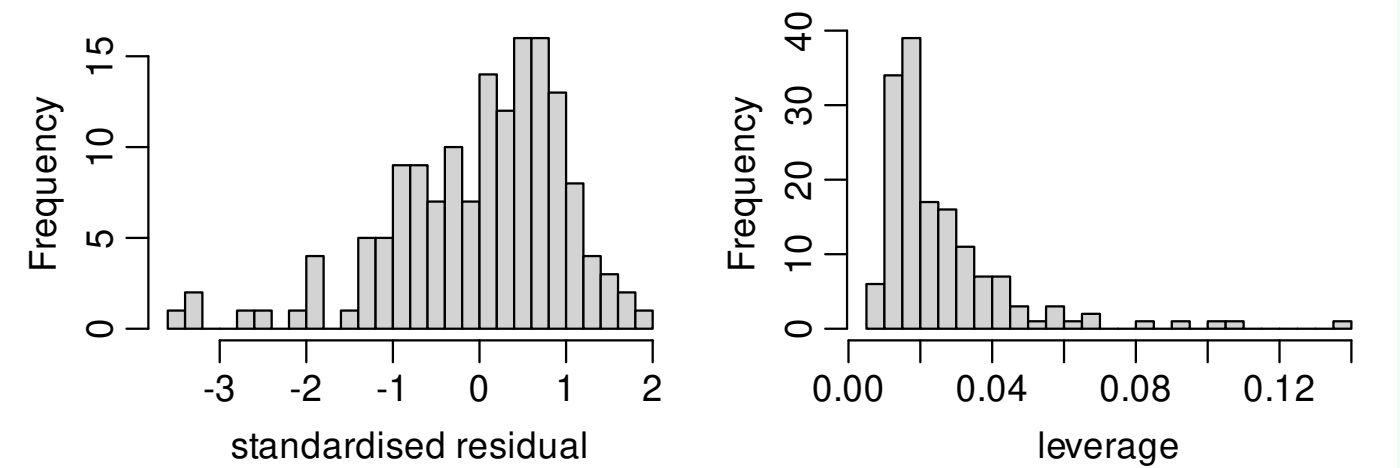
```
op <- par(mfrow = c(1, 4), mar = c(3.2, 2.8, 0.6, 0.4), mgp = c(1.9, 0.6, 0))
for (v in vs) plot(un[[v]], un$LIFEEXP, xlab = v, ylab = if (v == "GDP") "LIFEEXP"
 ↪  else "", pch = 16, cex = 0.4)
par(op)
```



`GDP`, in billions of dollars, has skewness 9.26: most countries are compressed against the left edge by a few large economies, and the correlation with life expectancy is 0.21. Its logarithm has skewness 0.23, spreads the countries evenly, and shows an increasing, nearly linear relation with correlation

0.48. Private health expenditure, a percentage of GDP, is right-skewed (1.16); its logarithm is mildly left-skewed (−0.685) and neither version is linearly related to life expectancy (correlations −0.04 and −0.03). The logarithm also reduces the leverage of the largest values (Proposition 4.12). (c)

```
d_un <- na.omit(un[, c("COUNTRY", "LIFEEXP", "FERTILITY", "PUBLICEDUCATION",
 ↪  "lnHEALTH")])
fit_un <- lm(LIFEEXP ~ FERTILITY + PUBLICEDUCATION + lnHEALTH, data = d_un)
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.4, 0.6, 0.6), mgp = c(1.9, 0.6, 0))
hist(rstandard(fit_un), breaks = 20, main = "", xlab = "standardised residual")
hist(hatvalues(fit_un), breaks = 20, main = "", xlab = "leverage")
par(op)
```




```
n_un <- nrow(d_un); i_les <- which(d_un$COUNTRY == "Lesotho")
c(n = n_un, r = rstandard(fit_un)[[i_les]], t = rstudent(fit_un)[[i_les]],
  h = hatvalues(fit_un)[[i_les]], cut = 3 * 4 / n_un, rank_h =
   ↪  rank(-hatvalues(fit_un))[[i_les]],
  D = cooks.distance(fit_un)[[i_les]], expected_flags = n_un * flag(n_un, 3),
  flags = sum(abs(rstandard(fit_un)) > 2))
```

```
##           n           r           t           h
##   152.00000    -3.24967    -3.36080     0.13923
```

```
##           cut         rank_h              D expected_flags
##       0.07895        1.00000        0.42703        6.85883
##         flags
##       6.00000
```

```
d_un$COUNTRY[order(-abs(rstandard(fit_un)))[1:6]]
```

```
## [1] "Zimbabwe"     "Swaziland"    "Lesotho"
## [4] "Botswana"     "South Africa" "Zambia"
```

```
fit_un2 <- update(fit_un, data = d_un[-i_les, ])
round(rbind(with = coef(summary(fit_un))[, 1], without = coef(summary(fit_un2))[,
    1],
            t_with = coef(summary(fit_un))[, 3], t_without =
                coef(summary(fit_un2))[, 3]), 3)
```

```
##           (Intercept) FERTILITY PUBLICEDUCATION lnHEALTH
## with            85.63    -5.399          -0.185   -1.030
## without         84.17    -5.308           0.140   -1.261
## t_with          42.74   -16.324          -0.688   -1.092
## t_without       42.39   -16.535           0.506   -1.379
```

```
round(c(R2_with = summary(fit_un)$r.squared, R2_without =
  summary(fit_un2)$r.squared,
        s_with = summary(fit_un)$sigma, s_without = summary(fit_un2)$sigma), 3)
```

```
##    R2_with R2_without     s_with  s_without
##      0.645      0.659      6.645      6.426
```

(i) On the 152 countries with complete data, the standardised residuals have a long lower tail, and the leverages are concentrated near their mean with a few large values. (ii) Lesotho has $r = -3.25$ ($t = -3.36$) and leverage 0.139, the largest of all and above $3(k+1)/n = 0.079$: it is both an outlier and a high leverage point, with Cook's distance 0.43. Six countries have $|r| > 2$, against 6.9 expected under a correct model (Proposition 4.7), and the five largest $|r|$ are all countries of southern Africa. (iii) Without Lesotho, the coefficient of `PUBLICEDUCATION` changes sign, from -0.185 to 0.140, significant in neither fit; `FERTILITY` moves from -5.40 to -5.31 and `lnHEALTH` from -1.03 to -1.26; R^2 rises from 0.645 to 0.659 and s falls from 6.65 to 6.43. ■

Exercise 4.4 (FREES exercise 5.4, parts (a)–(c))**.** The 275 term life purchasers: regress `LNFACE` on `LNINCOME`, `EDUCATION`, `NUMHH`, `MARSTAT`, `AGE` and `GENDER`. (a) Collinearity: (i) explain

collinearity and a variance inflation factor; (ii) what is a large one? (iii) what can be done about it? (iv) with a table of correlations, is collinearity an issue here? (b) Unusual points: (i) define leverage; (ii) give the rules of thumb and identify unusual points; (iii) how would the leverages change with **FACE** as the response? (c) Residual analysis: plot residuals against fitted values, a normal quantile plot, and residuals against leverages; what does each show?

Solution. MARSTAT enters as the indicators MAR1 (married) and MAR2 (living with a partner), as in Unit 3, so $k = 7$.

```
term <- subset(read.csv("data/frees/csv/TermLife.csv"), FACE > 0)
term$LNFACE <- log(term$FACE); term$LNINCOME <- log(term$INCOME)
term$MAR1 <- as.numeric(term$MARSTAT == 1); term$MAR2 <- as.numeric(term$MARSTAT ==
 ↪  2)
f6 <- lm(LNFACE ~ LNINCOME + EDUCATION + NUMHH + MAR1 + MAR2 + AGE + GENDER, data =
 ↪  term)
round(coef(summary(f6)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.8867     0.9308  3.1013   0.0021
## LNINCOME      0.4361     0.0790  5.5170   0.0000
```

```
## EDUCATION     0.2136     0.0387  5.5190   0.0000
## NUMHH         0.2342     0.0742  3.1555   0.0018
## MAR1          0.1791     0.3395  0.5276   0.5982
## MAR2         -0.6730     0.5757 -1.1690   0.2434
## AGE          -0.0059     0.0080 -0.7362   0.4623
## GENDER        0.7141     0.3726  1.9164   0.0564
```

```
round(car::vif(f6), 3)
```

```
##  LNINCOME EDUCATION     NUMHH      MAR1      MAR2       AGE
##     1.265     1.175     1.482     2.573     1.407     1.179
##    GENDER
##     1.959
```

```
v_term <- c("LNINCOME", "EDUCATION", "NUMHH", "MAR1", "MAR2", "AGE", "GENDER")
round(cor(term[v_term]), 3)
```

```
##           LNINCOME EDUCATION  NUMHH   MAR1   MAR2    AGE
## LNINCOME     1.000     0.343  0.179  0.280 -0.104  0.045
## EDUCATION    0.343     1.000 -0.064  0.054 -0.109  0.092
## NUMHH        0.179    -0.064  1.000  0.411  0.044 -0.318
## MAR1         0.280     0.054  0.411  1.000 -0.342  0.062
```

```
## MAR2        -0.104    -0.109  0.044 -0.342  1.000 -0.132
## AGE          0.045     0.092 -0.318  0.062 -0.132  1.000
## GENDER       0.229    -0.044  0.297  0.620  0.077  0.005
##            GENDER
## LNINCOME    0.229
## EDUCATION -0.044
## NUMHH       0.297
## MAR1        0.620
## MAR2        0.077
## AGE         0.005
## GENDER      1.000
```

```
n6 <- nobs(f6); h6 <- hatvalues(f6); n6
```

```
## [1] 275
```

```
c(cut_3 = 3 * 8 / n6, n_above = sum(h6 > 3 * 8 / n6), max_h = max(h6),
  max_D = max(cooks.distance(f6)), expected_flags = n6 * flag(n6, 7),
  flags = sum(abs(rstandard(f6)) > 2), min_r = min(rstandard(f6)), max_abs_t =
   ↪  max(abs(rstudent(f6))),
  bonferroni_cut = qt(1 - 0.05 / (2 * n6), n6 - 7 - 2))
```

```
##          cut_3        n_above          max_h          max_D
##        0.08727       17.00000        0.16776        0.14032
## expected_flags          flags          min_r      max_abs_t
##       12.45599       11.00000       -3.88574        3.99300
## bonferroni_cut
##        3.79650
```

```
term[names(which.max(h6)), c("INCOME", "EDUCATION", "NUMHH", "MARSTAT", "AGE",
  "GENDER", "FACE")]
```

```
##     INCOME EDUCATION NUMHH MARSTAT AGE GENDER FACE
## 427    500        14     2       2  20      1 2600
```

```
round((unlist(term[names(which.max(h6)), v_term]) - colMeans(term[v_term])) /
  sapply(term[v_term], sd), 2)
```

```
##  LNINCOME EDUCATION     NUMHH      MAR1      MAR2       AGE
##     -3.81     -0.21     -0.64     -1.76      5.14     -2.26
##    GENDER
##      0.39
```

```r
c(min_AGE = min(term$AGE), n_MARSTAT_2 = sum(term$MARSTAT == 2));
 ↪  sort(term$INCOME)[1:4]
```

```
##     min_AGE n_MARSTAT_2
##          20          10
## [1]  260  260  500 2000
```

```r
round(h6[term$INCOME == 260], 3)
```

```
##   360   393
## 0.119 0.107
```

```r
c(max_leverage_change_with_FACE = max(abs(hatvalues(update(f6, FACE ~ .)) - h6)))
```

```
## max_leverage_change_with_FACE
##                             0
```

```r
c(nR2 = unname(lmtest::bptest(f6)$statistic), p = lmtest::bptest(f6)$p.value)
```

```
##      nR2     p.BP
## 17.78788  0.01296
```

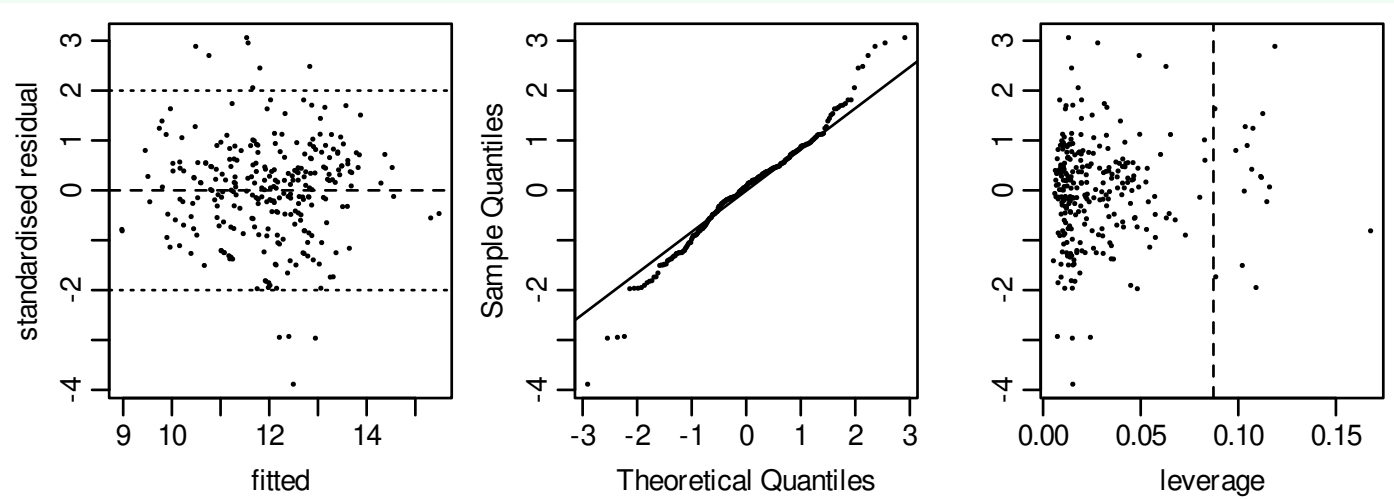


Figure 4.10: Standardised residuals against fitted values, their normal quantile plot, and against leverage, for the regression of `LNFACE` on seven variables.

(a)(i) Collinearity is a near-linear relation among the explanatory variables; $\text{VIF}_j = 1/(1 - R_j^2)$ is the factor by which it multiplies $\text{var}(\hat{\beta}_j)$ relative to uncorrelated variables (Unit 3's corollary **the standard error through the other explanatory variables** and Proposition 4.20). (ii)

FREES calls $\text{VIF}_j > 10$ severe, equivalently $R_j^2 > 0.9$; ISLR uses 5 or 10. (iii) Drop one of the related variables, combine them into one, replace one by a variable carrying similar information, or, for a polynomial, centre before squaring (Proposition 4.21). (iv) The largest factor is 2.57, for `MAR1`, and the largest correlation is 0.62, between `MAR1` and `GENDER`: collinearity is not what makes `MAR1`, `MAR2` and `AGE` insignificant.

(b)(i) The leverage h_{ii} is the i-th diagonal entry of $\mathbf{H}$, the weight of y_i in its own fitted value, and by Proposition 4.12 a function of the Mahalanobis distance of $\mathbf{x}_i$ from the centre of the design. (ii) FREES flags $h_{ii} > 3(k+1)/n = 0.087$; 17 households exceed it. The largest, 0.168, belongs to the youngest respondent, aged 20, one of the ten households with `MARSTAT` 2 and the third-lowest income, \$500: it lies 3.81 standard deviations below the mean in `LNINCOME`, 5.14 above in `MAR2` and 2.26 below in `AGE`, unusual jointly as Proposition 4.12 measures; the two households with income \$260 have leverages 0.119 and 0.107. No Cook's distance exceeds 0.15. (iii) Leverages depend on $\mathbf{X}$ alone, so with `FACE` as the response they are unchanged, as the zero difference confirms.

(c) Against the fitted values the standardised residuals show no trend; 11 exceed 2 in absolute value, against 12.5 expected (Proposition 4.7). The quantile plot follows the line except in the lower tail, where the smallest residual is -3.89; its studentised residual, 3.99 in absolute value,

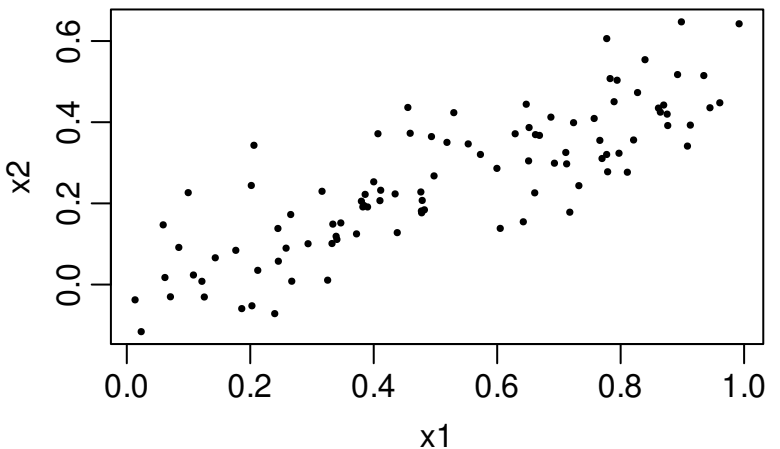
exceeds the Bonferroni cutoff 3.80 of Corollary 4.6. Koenker's statistic 17.8 on 7 degrees of freedom ($p = 0.013$) indicates some heteroscedasticity that the plot does not show. Against leverage, no point combines a large residual with a large leverage, as the small Cook's distances confirm. ■

Exercise 4.5 (ISLR §3.7, exercise 14)**.** (a) After `set.seed(1)`, generate x_1 uniform on $(0, 1)$, $x_2 = 0.5x_1 + \delta/10$ and $y = 2 + 2x_1 + 0.3x_2 + \varepsilon$ for 100 observations, with δ and ε standard normal (the commands open the solution); write the model and its coefficients. (b) What is the correlation of `x1` and `x2`? Plot them. (c) Regress `y` on both: compare the estimates with the true coefficients; can $H_0\colon \beta_1 = 0$ and $H_0\colon \beta_2 = 0$ be rejected? (d) Regress on `x1` only; (e) on `x2` only. (f) Do (c)–(e) contradict each other? (g) Add the mismeasured observation `x1` $= 0.1$, `x2` $= 0.8$, `y` $= 6$ and refit (c)–(e): is it an outlier, a high leverage point, or both, in each model?

Solution. (a) $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ with $\beta_0 = 2$, $\beta_1 = 2$, $\beta_2 = 0.3$ and $\varepsilon \sim N(0, 1)$.

```
1 set.seed(1)
2 x1 <- runif(100); x2 <- 0.5 * x1 + rnorm(100) / 10; y <- 2 + 2 * x1 + 0.3 * x2 +
   ↪  rnorm(100)
3 op <- par(mar = c(3.2, 3.4, 0.6, 0.6), mgp = c(1.9, 0.6, 0))
```

```r
plot(x1, x2, pch = 16, cex = 0.5)
par(op)
```




```r
c(cor_x1_x2 = cor(x1, x2), VIF = 1 / (1 - cor(x1, x2)^2))
```

```
## cor_x1_x2       VIF
##    0.8351    3.3050
```

```r
fits14 <- list(both = lm(y ~ x1 + x2), x1_only = lm(y ~ x1), x2_only = lm(y ~ x2))
lapply(fits14, function(m) round(coef(summary(m)), 4))
```

```
## $both
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.131     0.2319  9.1879   0.0000
## x1             1.440     0.7212  1.9961   0.0487
## x2             1.010     1.1337  0.8906   0.3754
##
## $x1_only
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.112     0.2307   9.155        0
## x1             1.976     0.3963   4.986        0
##
## $x2_only
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)     2.39     0.1949   12.26        0
## x2              2.90     0.6330    4.58        0
```

(b) The correlation is 0.835, and the scatter plot is an increasing band. (c) $\hat{\beta}_0 = 2.13$, $\hat{\beta}_1 = 1.44$ and $\hat{\beta}_2 = 1.01$: β_0 is well estimated, β_1 and β_2 poorly, with standard errors 0.72 and 1.13 inflated by VIF $= 3.30$ (Proposition 4.20). H_0: $\beta_1 = 0$ is rejected at 5%, barely ($p = 0.049$); H_0: $\beta_2 = 0$ is not ($p = 0.38$). (d) Alone, `x1` has slope 1.98 with $t = 4.99$, and (e) `x2` has slope 2.90 with

$t = 4.58$: both reject H_0. (f) No: the simple slope of `x2` estimates β_2 plus β_1 times the slope of `x1` on `x2` (Unit 3's omitted variable theorem), which is far from 0; and in (c) each variable is tested given the other, with standard errors inflated by the collinearity (Proposition 4.18), so each t ratio is small although the two jointly matter.

```
x1 <- c(x1, 0.1); x2 <- c(x2, 0.8); y <- c(y, 6)
fits14g <- list(both = lm(y ~ x1 + x2), x1_only = lm(y ~ x1), x2_only = lm(y ~ x2))
lapply(fits14g, function(m) round(coef(summary(m)), 4))
```

```
## $both
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.2267     0.2314  9.6244   0.0000
## x1            0.5394     0.5922  0.9109   0.3646
## x2            2.5146     0.8977  2.8012   0.0061
##
## $x1_only
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.257     0.2390   9.445        0
## x1             1.766     0.4124   4.282        0
##
```

```
## $x2_only
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.345     0.1912  12.264        0
## x2             3.119     0.6040   5.164        0
```

```
round(sapply(fits14g, function(m) c(t_101 = rstudent(m)[[101]], h_101 =
  hatvalues(m)[[101]],
  h_over_mean = hatvalues(m)[[101]] / (length(coef(m)) / 101),
  cut = 3 * length(coef(m)) / 101, D_101 = cooks.distance(m)[[101]])), 4)
```

```
##                both x1_only x2_only
## t_101        2.1135  3.4384  1.1410
## h_101        0.4147  0.0335  0.1013
## h_over_mean 13.9625  1.6903  5.1162
## cut          0.0891  0.0594  0.0594
## D_101        1.0190  0.1845  0.0732
```

```
c(cor_after = cor(x1, x2), VIF_after = 1 / (1 - cor(x1, x2)^2))
```

```
## cor_after VIF_after
##    0.7392    2.2049
```

(g) In the regression on both, the new point has leverage 0.415, fourteen times the mean and far

above $3(k+1)/n = 0.089$, and $t = 2.11$: a high leverage point and a mild outlier, with Cook's distance 1.02. It lies off the band of the other points (Proposition 4.12), lowers the correlation to 0.739, and reverses the conclusions of (c): $\hat{\beta}_1 = 0.54$ is no longer significant while $\hat{\beta}_2 = 2.51$ is ($p = 0.006$). On `x1` alone its leverage 0.033 is below 0.059 but $t = 3.44$: an outlier. On `x2` alone its leverage 0.101 exceeds 0.059 and $t = 1.14$: a high leverage point, not an outlier. ■

Exercise 4.6 (SOA sample question 14)**.** The spread of the residuals of an ordinary least squares fit increases with the predicted values, and the observed responses range from 0 to 100. Which transformations of the response may make the variance of the residuals more constant? I. The logarithm of one plus the response. II. The square root. III. The logit. (A) None; (B) I and II only; (C) I and III only; (D) II and III only; (E) the correct answer is not given by (A), (B), (C) or (D).

Solution. $g(y) = \ln(1+y)$ and $g(y) = \sqrt{y}$ are increasing and concave on $[0, \infty)$, with increasing convex inverses $h(t) = e^t - 1$ and $h(t) = t^2$ on $g([0, \infty))$. By Proposition 4.36, if $g(y_i)$ follows a linear model with identically distributed errors, the variance of y_i is nondecreasing in its mean, the pattern described, and the transformation returns the model to constant variance; so I and II

are true. $\ln(1+y)$ is defined at $y = 0$, where $\ln y$ is not. III is false: $\text{logit}(y) = \ln\{y/(1-y)\}$ is defined only for $0 < y < 1$, and even for $y/100$ it is infinite at the observed values 0 and 100. I and II only: (B). ■